Breaking Trivial Cognitive Frames

A Formal RAFASSABRA Method for Expanding the Space of Inquiry Through Interrogative Priority and Controlled Divergence

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Abstract

Every formal system begins with axioms—but before any axiom stands the question that motivated it. This paper proves that the space of formulable questions constrains the space of possible truths more fundamentally than axiomatic declaration, and presents a formal method for breaking through that constraint.

Three claims are established. **Claim A (Priority):** Questions are ontologically prior to answers. Under four independent formalisations of "question" (Hintikka partition semantics, Belnap–Steel erotetic implication, Hintikka's interrogative model, and quantum operator formalism) and three mathematical frameworks (set-theoretic, category-theoretic, information-theoretic), the question space is shown to carry strictly richer structure than the answer space—lattice ordering, non-faithful functors, complexity asymmetry—across five substantially independent proof paths expressed in three mathematical languages. **Claim B (Constraint):** Cognitive frames restrict the space of formulable questions more severely than they restrict available answers. This restriction asymmetry is proved formally and demonstrated computationally: a bounded agent necessarily occupies a frame that excludes a larger fraction of questions than of answers, and no increase in deductive power compensates for interrogative poverty. **Claim C (Method):** The RAFASSABRA (Random Angry Frozen Alien Stripped, Stole, Assembled, Broke, and Ran Away) method—a systematic method comprising structural attack, atemporal examination, presuppositional ascent, apophatic negation, cross-domain isomorphic transfer, cataphatic assertion, paradoxical stress-testing, and temporal reconstruction—provably expands the question space beyond the current frame. A Composition Theorem establishes that each stage expands in a direction unique to that stage, and that no stage is redundant.

The method incorporates an externally-sourced random perturbation at every stage, formalised both as orthogonal projection in a high-dimensional concept space and as mutation in an evolutionary search over the partition lattice, with proof that both formalisations yield the same expansion guarantee. Each result is proved, then subjected to attempted counter-proof whose failure is analysed. The method is demonstrated empirically on five substantive problems (the nature of time, consciousness, philosophy, corruption, and dark matter) with independent parallel runs and convergence analysis. Structural features converge across runs; expressive features diverge—confirming that the method's outputs are properties of the problem, not of the executor.

Roadmap

The paper is organised in three Parts. **Part I** (Sections 1–12) presents the RAFASSABRA method: its nine stages, formal properties, expansion theorems, and empirical demonstrations. **Part II** (Sections 13–17) proves the structural priority of questions over answers under four formalisations and three mathematical frameworks. **Part III** (Section 18) proves that cognitive frames restrict questions more severely than answers. This ordering places the method—the paper's most distinctive contribution—first. Readers seeking the theoretical foundations before the method may begin with Part II, then Part III, then return to Part I.

Contents

Part I
The RAFASSABRA Method

1 From Priority to Method

If questions are ontologically prior to answers—if the space of formulable questions constrains the space of possible truths more fundamentally than axiomatic declaration—then the primary obstacle to knowledge is not the difficulty of finding answers but the difficulty of formulating questions. And if cognitive frames restrict the space of formulable questions more severely than they restrict available answers, then no amount of deductive or experimental power compensates for the inability to ask the right question.

This Part presents a systematic method for expanding the space of formulable questions beyond the current cognitive frame. The method—the **RAFASSABRA (Random Angry Frozen Alien Stripped, Stole, Assembled, Broke, and Ran Away) method**—operates through a preliminary structural attack, an atemporal foundation pass, and six stages of structured questioning, with an externally-sourced random perturbation at each stage.

The stages are:

Stage 1. Random Words (*Random*). Before any analysis, obtain eight random words from an external source (e.g., `randomwordgenerator.com`). The words must be generated by someone who does not know your topic. Do not generate your own—self-generated words stay within your frame (Theorem 10.3). One word will be used at each of Stages 2–9.

Stage 2. Structural Attack (*Angry*). Attempt to invalidate the question before analysis begins.

Stage 3. Atemporal Pass (*Frozen*). Remove time; reconstruct the phenomenon as static relations.

Stage 4. Structural Necessity / Tree-Top Ascent (*Alien*). Identify conditions required for existence by iterative presupposition.

Stage 5. Irreducible Residue / Apophatic Negation (*Stripped*). Strip all removable features; retain only what survives total negation.

Stage 6. Isomorphic Transfer (*Stole*). Find the same formal structure in at least three unrelated domains; import their questions.

Stage 7. Cataphatic Assertion (*Assembled*). Construct a positive model and derive its predictive consequences.

Stage 8. Paradoxical Stress-Testing (*Broke*). Search the model for internal contradictions; generate questions from each.

Stage 9. Temporal Reconstruction (*Ran Away*). Reintroduce time; distinguish structural base from temporal projection.

1.1 Why Random Words Come First

Stage 1 is placed before all analytical work for a reason that is both practical and theoretical.

Practical: if the inquirer encounters the random words after beginning analysis, the frame has already formed. The word "refrigerator," encountered after twenty minutes of thinking about corruption, will be assimilated into the corruption frame—interpreted as metaphor rather than structural constraint. The same word, encountered before any analysis, cannot be assimilated because there is no frame yet to assimilate it into. The words must be alien not only to the topic but to the inquirer's current state of engagement with the topic.

Theoretical: the random word mechanism (Section 10) guarantees frame-externality only if the words are independent of the frame. Independence requires that word selection precedes frame activation. If the inquirer first formulates the question (activating the frame) and then obtains random words, the inquirer's selection of a random word generator, interpretation of the words, and even unconscious filtering of which words to "really use" will be contaminated by the active frame. Stage 1 enforces temporal separation between word acquisition and frame engagement.

Empirical: the control experiment (Section 12) demonstrated that self-generated "random" words are absorbed by the frame without residue ($\|\vec{w}_\perp\| \approx 0$). Externally-sourced words, obtained before analysis, produced structural deformation in 10 of 12 cases. The ordering matters: randomness must precede reasoning, or it ceases to be random.

1.2 Formal Setup

Let W be a logical space (a set of possible worlds). Let $\Pi(W)$ denote the partition lattice on W—the set of all questions formulable about W, ordered by refinement. A **cognitive frame** $\mathfrak{F}$ is a sublattice $\mathfrak{F} \subseteq \Pi(W)$: a coherent collection of questions closed under meet and join.

An agent operating within $\mathfrak{F}$ can formulate only questions in $\mathfrak{F}$. The goal of the method is to produce a strictly larger frame $\mathfrak{F}' \supsetneq \mathfrak{F}$.

Each stage S_k is an operator on frames:

$$\mathfrak{F}_{\text{init}} \xrightarrow{S_1} \mathfrak{F}^+_{\text{init}} \xrightarrow{S_2} \mathfrak{F}^+_{\text{init}} \xrightarrow{S_3} \mathfrak{F}_3 \xrightarrow{S_4} \mathfrak{F}_4 \xrightarrow{S_5} \mathfrak{F}_5 \xrightarrow{S_6} \mathfrak{F}_6 \xrightarrow{S_7} \mathfrak{F}_7 \xrightarrow{S_8} \mathfrak{F}_8 \xrightarrow{S_9} \mathfrak{F}_9.$$

The central claim is that $\mathfrak{F}_9 \supsetneq \mathfrak{F}_{\text{init}}$ under generic conditions, with each inclusion strict (Theorem 11.1).

1.3 Mnemonic: The RAFASSABRA method

The stages are encoded in a mnemonic:

"Random Angry Frozen Alien Stripped, Stole, Assembled, Broke, and Ran Away."

The phrase begins with *Random*—and this is not an accident. The random word mechanism is the method's most distinctive and most counterintuitive feature: at every stage, an externally-sourced random word must be incorporated as a structural constraint. The word "Random" comes first because randomness *precedes* the method. Without external perturbation, every stage operates within the current frame; with it, every stage is forced outside. The entire method is a scaffold for making randomness productive rather than destructive (Section 10).

Notice, too, that the mnemonic phrase itself is random. "Random Angry Frozen Alien Stripped, Stole, Assembled, Broke, and Ran Away" is not a sentence anyone would construct deliberately. It is absurd, incongruous, and memorable precisely because it is alien to ordinary discourse—which is exactly how an externally-sourced random word functions within the method. The mnemonic is a demonstration of its own principle: a random phrase, forcibly assembled into a coherent structure, produces something that a deliberate phrase would not. You will not forget the angry frozen alien.

It is advisable to visualise the scene: a *random* word falls from the sky onto a stage where an *angry* alien—visibly furious, fists clenched—stands *frozen* in a block of ice. The ice cracks. The alien *strips* off its clothes, piece by piece, down to its irreducible core. Naked and shivering, it *steals* a coat from a passing physicist, a hat from a biologist, boots from an economist. Thus *assembled* in a patchwork outfit, it admires itself—then paradoxically *breaks* the mirror. And before anyone can react, it *runs away*, back into the flow of time, wearing a new frame.

Random → obtain eight random words (Stage 1). **A**ngry → Attack the question (Stage 2). **F**rozen → Freeze time (Stage 3). **A**lien → Ascend presuppositions (Stage 4). **S**tripped → Strip to irreducible residue (Stage 5). **S**tole → Steal structure across domains (Stage 6). **A**ssembled → Assert a model (Stage 7). **B**roke → Break the model (Stage 8). **Ran Away** → Restore time and escape the old frame (Stage 9).

2 Stage 2: Structural Attack

2.1 Informal Description

Before the method begins its constructive work, it attempts to destroy its own object. The inquirer subjects the phenomenon or question to a structural attack: an attempt to show that the question is incoherent, that the phenomenon does not exist as a distinct entity, or that hidden presuppositions render the inquiry ill-formed.

The attack has four components: (i) attempt to derive a contradiction from the question's presuppositions; (ii) identify presuppositions that the questioner has not made explicit; (iii) test whether the phenomenon named by the question exists as a coherent object or is a conflation of distinct objects; (iv) if a contradiction or incoherence is found, state it precisely and halt.

The halt condition is strict: the attack must derive a *structural* impossibility, not merely express doubt. Casual rejection is forbidden. The question is presumed coherent until proven otherwise.

2.2 Formalisation

Definition 2.1 (Well-Formedness Test)**.** A question Q with direct answers $\mathrm{dQ} = \{\alpha_1, \ldots, \alpha_k\}$ and presuppositions $P = \mathrm{presup}(Q)$ passes the **well-formedness test** if and only if:

(i) P is consistent: $P \nvdash \bot$.

(ii) $|\mathrm{dQ}| \geq 2$: at least two distinct direct answers.

(iii) $P \vdash \bigvee_i \alpha_i$: the question is sound.

(iv) $P \nvdash \alpha_i$ for any individual i: the question is open.

Theorem 2.2 (Attack Soundness)**.** *The structural attack operator rejects Q_{init} only if Q_{init} is structurally defective. No well-formed question is rejected.*

Proof. The operator tests the four conditions of Definition 2.1. Each condition is decidable for finite W. The operator rejects if and only if at least one condition fails. Since well-formed questions satisfy all four conditions by definition, no well-formed question is rejected. □

Why this step matters

The structural attack saves effort (early detection of incoherent questions) and forces the inquirer to make presuppositions explicit. The question "Why is the sky blue?" presupposes that the sky is blue, that "blue" is a property of the sky rather than of the observer, and that "why" requests a causal rather than a functional explanation. Making these presuppositions explicit sharpens the question before analysis begins.

Self-Critique and Resolution

Residual concern. The structural attack assumes that the inquirer can identify all presuppositions. But presuppositions are often invisible—they are the water the fish does not see.
Resolution. This limitation is partially addressed by Stage 4 (ascent), which systematically uncovers presuppositions by iterative "why" questioning. The structural attack catches obvious defects; the ascent catches deep ones. The two are complementary.

Example of a productive halt. When the method was applied to "the meaning of life," Stage 2 flagged a hidden presupposition: "meaning" presupposes an external evaluator. The halt itself was the finding—the question is structurally defective unless the evaluator is specified. Reformulating as "what structural features distinguish states that agents persistently seek from states they avoid?" survived the attack and produced a productive analysis. The result is definitional but serves to establish that the attack operator has no false positives—it rejects only questions whose presuppositions are genuinely incoherent.

3 Stage 3: Atemporal Pass

3.1 Informal Description

Before constructing any model, the inquirer removes time entirely. No sequence, no change, no causation, no "before" and "after." The phenomenon is reconstructed as a set of static relations: a geometry without history.

If the phenomenon collapses without time, temporality is *irreducible*. If it survives, time is a *projection*: a way of viewing that adds convenience but not content.

3.2 Formalisation

Definition 3.1 (Temporal and Atemporal Questions)**.** Let W be a logical space with temporal structure $\tau : W \to W$. A question $Q \in \Pi(W)$ is **temporally dependent** if $\tau(\pi_Q) \neq \pi_Q$. A question is **atemporal** if $\tau(\pi_Q) = \pi_Q$.

Definition 3.2 (Atemporal Projection)**.** For a temporally dependent question Q^t, the **atemporal projection** Q^a is induced by the coarsest partition agreeing with π_{Q^t} on all time-invariant properties: $\pi_{Q^a} = \{[B]_\tau : B \in \pi_{Q^t}\}$, where $[B]_\tau$ is the equivalence class of B under the relation $w \sim_\tau w'$ iff w and w' differ only in temporal properties.

Definition 3.3 (Atemporal Operator)**.** S_0 maps $\mathfrak{F}_{\text{init}}$ to $\mathfrak{F}_0 = \mathfrak{F}_{\text{init}} \cup \{Q^a : Q^t \in \mathfrak{F}_{\text{init}}\} \cup \{Q_{\text{irred}}\}$, where Q_{irred} asks whether temporality is irreducible or projective.

Theorem 3.4 (Atemporal Expansion)**.** *If $\mathfrak{F}_{\text{init}}$ contains at least one temporally dependent question, then $\mathfrak{F}_0 \supsetneq \mathfrak{F}_{\text{init}}$.*

Proof. Let $Q^t \in \mathfrak{F}_{\text{init}}$ be temporally dependent. Its atemporal projection Q^a has a coarser partition (merging blocks distinguished only by temporal properties). Since Q^t is temporally dependent, $\pi_{Q^a} \neq \pi_{Q^t}$, so Q^a is distinct from Q^t. Generically $Q^a \notin \mathfrak{F}_{\text{init}}$ because the initial frame defaults to temporal framing. Therefore $Q^a \in \mathfrak{F}_0 \setminus \mathfrak{F}_{\text{init}}$. □

Why this step matters

The atemporal pass is placed at Stage 3 because its output determines the vocabulary for all subsequent stages. Some of the deepest results in physics emerged from atemporal reformulations: the block universe of special relativity, the Wheeler–DeWitt equation, the path integral formulation. Removing time reveals structure that temporal framing obscures.

Self-Critique and Resolution

Residual concern. Temporal reasoning is so deeply embedded in natural language that truly atemporal description may be impossible for human inquirers.

Resolution. The pass aims not for perfection but for displacement: forcing the inquirer to notice how much of their reasoning is temporal and to ask whether each temporal element is essential. Even imperfect partial removal generates questions that a fully temporal approach would not produce.

4 Stage 4: Tree-Top Ascent

4.1 Informal Description

The inquirer begins with a phenomenon and asks: "What must be true for this to exist at all?" The answer identifies a presupposition. The inquirer then asks the same question of the presupposition. This continues until no new presupposition emerges—the limit of formal generalisation Kriger (2026).

4.2 Formalisation

Definition 4.1 (Presupposition Chain). Given $Q_0 \in \mathfrak{F}_0$, the **presupposition chain** is the sequence $Q_0, Q_1, Q_2, \ldots$ where Q_{i+1} is evoked by the presuppositions of Q_i. The chain terminates when no new question is evoked.

Definition 4.2 (Ascent Operator). S_1 maps $\mathfrak{F}_0$ to $\mathfrak{F}_1 = \mathfrak{F}_0 \cup \bigcup_{Q_0 \in \mathfrak{F}_0} \mathrm{Chain}(Q_0)$.

Theorem 4.3 (Ascent Expansion). *If $\mathfrak{F}_0$ contains at least one question with non-trivial presuppositions, then $\mathfrak{F}_1 \supsetneq \mathfrak{F}_0$.*

Proof. Let $Q_0 \in \mathfrak{F}_0$ with presuppositions P such that $\mathrm{Ev}(P, Q_1)$ for some $Q_1 \notin \mathfrak{F}_0$. Then $Q_1 \in \mathrm{Chain}(Q_0) \subseteq \mathfrak{F}_1$, so $Q_1 \in \mathfrak{F}_1 \setminus \mathfrak{F}_0$. □

Why this step matters

The ascent expands the frame vertically—toward more abstract questions. Cognitive frames tend to be rich at one level of expertise and sparse at higher abstraction levels. A biologist may have hundreds of questions about cellular mechanisms but none about what "mechanism" itself presupposes.

> ***Counter-Argument.*** *Presupposition chains are deterministic: $\mathfrak{F}_1$ is a deterministic function of $\mathfrak{F}_0$. No genuine expansion has occurred; latent content is merely unpacked.*

For bounded agents, the distinction between implicit and explicit is the distinction between reachable and unreachable. The counter-argument proves too much: by the same logic, every theorem of mathematics is "implicit" in the axioms, rendering all mathematics trivially known.

Self-Critique and Resolution

Residual concern. The ascent produces only questions logically entailed by presuppositions. It cannot generate logically independent questions.
Resolution. This is why the method has five additional stages. The ascent expands vertically; horizontal expansion requires isomorphism (Stage 6) and the random word mechanism.

5 Stage 5: Apophatic Negation

5.1 Informal Description

The inquirer systematically strips features from the phenomenon. For each feature: "Does the phenomenon survive without this?" What remains after all removable features are stripped is the *irreducible residue*.

The most radical negation targets the phenomenon itself: "Is this a genuine entity, or an explanatorily redundant label?" This connects to the Functional Sufficiency Framework Kriger (2025).

5.2 Formalisation

Definition 5.1 (Anti-Question)**.** For each feature α_i of question Q, the anti-question $\bar{Q}_i$ is induced by restricting π_Q to $W \setminus B_i$: $\pi_{\bar{Q}_i} = \{B_j \cap (W \setminus B_i) : j \neq i,\ B_j \cap (W \setminus B_i) \neq \emptyset\}$.

Definition 5.2 (Negation Operator)**.** S_2 maps $\mathfrak{F}_1$ to $\mathfrak{F}_2 = \mathfrak{F}_1 \cup \{\bar{Q}_i : Q \in \mathfrak{F}_1, \alpha_i \in \mathrm{dQ}\} \cup \{Q_{\text{exist}}\}$.

Theorem 5.3 (Negation Expansion)**.** *If any $Q \in \mathfrak{F}_1$ has $|\mathrm{dQ}| \geq 3$, then $\mathfrak{F}_2 \supsetneq \mathfrak{F}_1$.*

Proof. Let $Q \in \mathfrak{F}_1$ with $k \geq 3$ blocks. The anti-question $\bar{Q}_1$ is defined on the restricted space $W \setminus B_1$. No operation in the ascent produces domain-restricted questions, so $\bar{Q}_1 \notin \mathfrak{F}_1$. Therefore $\bar{Q}_1 \in \mathfrak{F}_2 \setminus \mathfrak{F}_1$. □

Why this step matters

Historically, the most productive scientific questions have been anti-questions. "What is motion without friction?" (Newton). "What is geometry without the parallel postulate?" (non-Euclidean geometry). "What is electrodynamics without the ether?" (Einstein). Removing taken-for-granted features revealed invisible structure.

Self-Critique and Resolution

Residual concern. The self-negation question risks nihilism.
Resolution. Most fundamental entities survive self-negation (mass, charge, logical implication). The self-negation is a filter separating load-bearing structures from decorative labels. Stage 7 (assertion) rebuilds from the surviving residue.

6 Stage 6: Isomorphic Transfer

6.1 Informal Description

The inquirer asks: where else does this structure appear? If the same formal structure governs phenomena in unrelated domains, then questions from those domains can be imported—not as metaphors but as formally identical questions. The criterion is strict: mathematical isomorphism, not verbal analogy.

6.2 Formalisation

Definition 6.1 (Structural Isomorphism)**.** Questions $Q \in \Pi(W)$ and $Q' \in \Pi(W')$ on different logical spaces are **structurally isomorphic** ($Q \cong_s Q'$) if there exist (i) a bijection $\phi : \pi_Q \to \pi_{Q'}$ and (ii) an extension to a lattice isomorphism of local neighbourhoods preserving refinement.

Definition 6.2 (Isomorphism Operator)**.** S_3 identifies structurally isomorphic questions across domains and imports their extended neighbourhoods: $\mathfrak{F}_3 = \mathfrak{F}_2 \cup \bigcup \Phi_j^{-1}(\mathcal{N}(Q'_j) \setminus \Phi_j(\mathcal{N}(Q) \cap \mathfrak{F}_2))$.

Theorem 6.3 (Isomorphism Expansion)**.** *If a cross-domain structural isomorphism exists with refinements not in $\mathfrak{F}_2$, then $\mathfrak{F}_3 \supsetneq \mathfrak{F}_2$.*

Proof. The refinements of Q'_j in $\Pi(W_j)$, pulled back through Φ_j^{-1}, yield questions not in $\mathfrak{F}_2$ by hypothesis. □

Why this step matters

The isomorphism operator provides horizontal expansion. Maxwell's equations were imported from fluid dynamics. Shannon's entropy from thermodynamics. Black–Scholes from the heat equation. Each was a structural isomorphism, not a metaphor.

Self-Critique and Resolution

Residual concern. Not all isomorphic transfers produce meaningful questions.
Resolution. The method requires minimum three independent matches (convergence filter). The isomorphism stage is generative; evaluation is deferred to Stages 7–9. The prompt specifies: "Same mathematical object, not analogy."

7 Stage 7: Cataphatic Assertion

7.1 Informal Description

Stages 4–6 disassemble the phenomenon. Stage 7 reverses direction: the inquirer constructs a positive model from the surviving material. The model is a set of committed claims about what the phenomenon *is*. Its purpose is not final truth but to create a target for Stages 8–9.

7.2 Formalisation

Definition 7.1 (Model)**.** A **model** M relative to $\mathfrak{F}_3$ is a consistent subset $M \subseteq \mathfrak{F}_3$ with a selection function $\sigma : M \to \bigcup_{Q \in M} \mathrm{d}Q$ choosing one direct answer per question. The commitment set is $\Gamma_M = \{\sigma(Q) : Q \in M\}$.

Definition 7.2 (Assertion Operator)**.** S_4 maps $\mathfrak{F}_3$ to $\mathfrak{F}_4 = \mathfrak{F}_3 \cup \mathrm{Pred}(M)$, where $\mathrm{Pred}(M) = \{Q : \mathrm{Ev}(\mathrm{Cn}(\Gamma_M), Q)\}$ are the predictive questions evoked by the model's deductive closure.

Theorem 7.3 (Assertion Expansion). *If* $\mathrm{Cn}(\Gamma_M)$ *contains a disjunction* $\bigvee_i \alpha_i$ *with no individual* α_i *derivable, then* $\mathfrak{F}_4 \supsetneq \mathfrak{F}_3$.

Proof. The disjunction evokes a predictive question Q_{pred} (sound and open relative to Γ_M). Since $\mathfrak{F}_3$ was constructed without committing to specific answers, $Q_{\text{pred}} \notin \mathfrak{F}_3$ generically. □

Why this step matters

The assertion is placed fourth, not first, because premature assertion restricts the question space to the neighbourhood of the initial guess. An agent who defers assertion until after three stages of frame expansion builds a model from $\mathfrak{F}_3 \supsetneq \mathfrak{F}_0$—a frame enriched by structural necessities, residues, and cross-domain imports. The deferral is a safeguard against premature closure.

Self-Critique and Resolution

Residual concern. If the selected answers are wrong, the predictive questions will be misleading.
Resolution. By design. False models can be extraordinarily productive: Bohr's model was false but generated exactly the right questions. The model is a scaffold, not a foundation. Stage 8 stress-tests it.

8 Stage 8: Paradoxical Stress-Testing

8.1 Informal Description

The model from Stage 7 is subjected to deliberate destruction. The inquirer searches for internal contradictions and treats them as *interrogative resources*: each contradiction generates a question the agent would not otherwise have asked.

8.2 Formalisation

Definition 8.1 (Contradiction Set). $\mathrm{Contra}(M) = \{(\alpha, \neg\alpha) : \alpha \in \mathrm{Cn}(\Gamma_M),\ \neg\alpha \in \mathrm{Cn}(\Gamma_M)\}$.

Definition 8.2 (Paradox Operator). S_5 maps $\mathfrak{F}_4$ to $\mathfrak{F}_5 = \mathfrak{F}_4 \cup \{Q_{\text{res}} : (\alpha, \neg\alpha) \in \mathrm{Contra}(M)\} \cup \{Q_{\text{level}}\}$, where each Q_{res} asks which premises are responsible for the contradiction, and Q_{level} asks whether the model conflates levels of description.

Theorem 8.3 (Paradox Expansion). *If* $\mathrm{Contra}(M) \neq \emptyset$*, then* $\mathfrak{F}_5 \supsetneq \mathfrak{F}_4$.

Proof. Resolution questions are meta-questions about derivational structure, not generated by any previous stage. Therefore $Q_{\text{res}} \notin \mathfrak{F}_4$. □

Why this step matters

The paradox stage inverts standard logic. Contradictions are not failures but signals marking the precise location where assumptions clash. The contradiction between Maxwell's equations and Galilean invariance produced special relativity. The contradiction between equipartition and observed specific heats produced quantum mechanics.

Self-Critique and Resolution

Residual concern. In tightly coupled models, localisation of contradiction may be impossible.
Resolution. This reveals tight coupling itself—a structural discovery generating the question: "Is the coupling genuine or artefactual?" Even the failure of localisation produces a question.

9 Stage 9: Temporal Reconstruction

9.1 Informal Description

Stage 3 removed time. Stage 9 reintroduces it deliberately, asking which elements require temporal expression and which do not. The result is a separation into a *structural base* (atemporal features) and a *temporal projection* (dynamic features that supervene on the base).

9.2 Formalisation

Definition 9.1 (Temporal Reconstruction Map)**.** The reconstruction map ρ associates to each atemporal question Q^a the set of temporally dependent questions that project onto it: $\rho(Q^a) = \{Q^t \in \mathfrak{F}_5 : \text{atemporal projection of } Q^t \text{ is } Q^a\}$.

Definition 9.2 (Separation Question)**.** For each Q^a with $|\rho(Q^a)| > 1$, Q_{sep} asks what determines the choice among temporal realisations.

Definition 9.3 (Temporal Reconstruction Operator)**.** S_6 maps $\mathfrak{F}_5$ to $\mathfrak{F}_6 = \mathfrak{F}_5 \cup \{Q_{\text{sep}} : |\rho(Q^a)| > 1\} \cup \{Q_{\text{nec}}\}$.

Theorem 9.4 (Temporal Reconstruction Expansion)**.** *If $\mathfrak{F}_5$ contains at least one atemporal question with $|\rho(Q^a)| > 1$, then $\mathfrak{F}_6 \supsetneq \mathfrak{F}_5$.*

Proof. The separation question asks about the relationship between atemporal structure and temporal realisation—a meta-structural question not generated by any previous stage. $\square$

Why this step matters

In physics, the structural base corresponds to conservation laws; temporal projections to dynamical trajectories. The laws are necessary; trajectories are contingent on initial conditions. The temporal reconstruction reveals which features are robust (structural) and which are fragile (projective).

Self-Critique and Resolution

Residual concern. The separation may not be unique—different choices of temporal structure τ may yield different separations.
Resolution. This non-uniqueness is a feature. The method can be run with multiple τ, and the intersection of structural bases identifies features that are atemporal under all framings—the truly time-independent core. This mirrors coordinate invariance in physics.

10 The Random Word Mechanism

Each stage receives an externally-sourced random word. The word is a structural constraint: the inquirer must incorporate it in a way that deforms the ontology of the phenomenon.

10.1 Formalisation 1: Orthogonal Projection (Motivating Model)

The following result is presented as a motivating model that captures the geometric intuition of the random word mechanism. The assumptions (uniform distribution on a linear subspace) are idealisations; real concept spaces have non-uniform geometry. The qualitative prediction—that random words are nearly maximally alien to any fixed frame—is supported empirically (Section 12) and by the weaker Formalisation 2.

Definition 10.1 (Concept Space)**.** A **concept space** $\mathcal{C}$ is a real vector space of dimension D whose basis vectors correspond to primitive concepts Mikolov et al. (2013). A frame $\mathfrak{F}$ occupies a subspace $V_{\mathfrak{F}} \subseteq \mathcal{C}$ of dimension $d \ll D$.

Definition 10.2 (Frame-Orthogonal Component)**.** For word $\vec{w} \in \mathcal{C}$: $\vec{w}_{\parallel} = \mathrm{proj}_{V_{\mathfrak{F}}}(\vec{w})$, $\vec{w}_{\perp} = \vec{w} - \vec{w}_{\parallel}$. The word is **frame-alien** if $\|\vec{w}_{\perp}\| > 0$.

Theorem 10.3 (Orthogonal Expansion Guarantee)**.** *For a random unit vector in $\mathbb{R}^D$: $\mathbb{E}[\|\vec{w}_{\perp}\|^2] = (D-d)/D$. For $D \sim 10^4$, $d \sim 10^2$, this exceeds 0.99. The random word is almost entirely outside the frame.*

Proof. A uniform random unit vector in $\mathbb{R}^D$ is isotropic: its expected squared projection onto any d-dimensional subspace is d/D. Since $V_{\mathfrak{F}}$ has dimension d, $\mathbb{E}[\|\vec{w}_{\parallel}\|^2] = d/D$ and $\mathbb{E}[\|\vec{w}_{\perp}\|^2] = 1 - d/D = (D-d)/D$. Furthermore, $\|\vec{w}_{\perp}\| = 0$ only if $\vec{w} \in V_{\mathfrak{F}}$, a set of measure zero when $d < D$. □

10.2 Formalisation 2: Evolutionary Mutation

Definition 10.4 (Mutation Operator)**.** A **mutation** of question Q by random word w is $Q' = Q \wedge \pi_w$ (meet in the partition lattice), where π_w is the partition induced by w.

Theorem 10.5 (Mutation Escapes Local Optima)**.** *Without mutation, a question population under selection and crossover converges to a local optimum within $\langle \mathcal{P}_0 \rangle \subseteq \mathfrak{F}_0$. With mutation by externally-sourced words, the population escapes with positive probability.*

Proof. Selection and crossover are closed on the sublattice $\langle \mathcal{P}_0 \rangle$. The word w induces $\pi_w \notin \mathfrak{F}_0$ with probability 1 (by Theorem 10.3). The mutant $Q' = Q \wedge \pi_w$ generically escapes $\langle \mathcal{P}_0 \rangle$. By Borel–Cantelli, beneficial mutations occur with probability 1 over infinitely many attempts Holland (1975). □

Why this step matters

Both formalisations predict the same outcome: externally-sourced random words expand the question space with probability approaching 1 as D/d grows. The convergence mirrors the paper's strategy: proving the same result in multiple frameworks establishes it as structural fact.

10.3 Bridging the Two Formalisations

The geometric model (Formalisation 1) predicts near-maximal frame-externality under idealised assumptions: uniform distribution, linear subspaces, high dimensionality. The evolutionary model (Formalisation 2) guarantees escape from local optima under much weaker assumptions but without quantitative force. The empirical observation (10/12 productive deformations in the control experiment, Section 12) is consistent with both but licensed by neither alone. The practical claim is therefore modest: externally-sourced random words have a non-negligible probability of productive structural deformation, and this probability is sufficient—over eight words per inquiry—to make at least some genuine expansion highly likely. The two formalisations bracket the truth from above (geometric idealisation) and below (evolutionary minimum), and the empirical rate falls between them.

Self-Critique and Resolution

Residual concern. The success rate of useful perturbations may be low.
Resolution. Empirically, 10/12 externally-sourced words produced retained deformations (Section 12). The blessing of dimensionality explains this: in high dimensions, random vectors are nearly maximally orthogonal to any fixed subspace.

10.4 The Role of Controlled Cognitive Divergence in Breaking Structural Persistence

A critical, albeit unconventional, component of the RAFASSABRA method is its deliberate mimicry of what might be termed "controlled cognitive divergence." In classical cognitive science, genius is often observed at the boundary between rigid logical persistence and the fluid, hyper-associative states typically associated with schizotypy.

The Random Word Anchor Rule serves as a formal heuristic to induce this state. By forcing the integration of ontologically alien concepts (e.g., "refrigerator" into "corruption"), the engine artificially suspends the fences of normative category boundaries. This process achieves two things.

First, *destabilisation of the common-sense frame*: it bypasses the evolutionarily derived persistence of standard mental models that filter out "impossible" or "illogical" questions. The common-sense frame is not neutral—it is a selection pressure that eliminates questions before they can be formulated, on the grounds that they "do not make sense." The random word overrides this filter by making the nonsensical mandatory.

Second, *access to latent isomorphisms*: it allows the system to identify structural similarities across domains that a purely convergent logic would reject as noise. When the inquirer is forced to connect "refrigerator" to "corruption," the connection cannot be superficial (the word is too alien for a surface match). The inquirer must reach for deep structure—and in reaching, may discover that corruption preserves (like a refrigerator preserves) informal equilibria that would otherwise decay. This is a genuine structural isomorphism, invisible from within the standard anti-corruption frame.

From a structural systems perspective, what is colloquially called "genius" is the ability of a system to maintain coherence while undergoing a radical shift in its foundational constraints. The Engine formalises this not as a loss of reason but as a temporary, high-energy departure from the path of least resistance. The mathematical guarantee (Theorem 10.3) ensures that the departure is genuine (the random word is almost maximally orthogonal to the frame), while the six-stage structure ensures that the departure is controlled (the stages evaluate, filter, and integrate the perturbation rather than allowing it to destabilise the inquiry permanently).

This suggests that the expansion of the question space requires a calculated risk: the willingness to allow a system to become "meaningfully unstable" before re-crystallising into a new, more expansive model. The method provides the scaffold for this instability—the structured sequence of stages that holds the inquiry together during the transition from one frame to another. Without the scaffold, the perturbation is noise. With it, the perturbation is expansion.

11 Composition Theorem

Theorem 11.1 (Strict Expansion of the Full Method)**.** *Under generic conditions, the composition* $S_6 \circ S_5 \circ S_4 \circ S_3 \circ S_2 \circ S_1 \circ S_0 \circ S_{\mathrm{pre}}$ *yields:* $\mathfrak{F}_{\mathrm{init}} \subsetneq \mathfrak{F}_3 \subsetneq \mathfrak{F}_4 \subsetneq \mathfrak{F}_5 \subsetneq \mathfrak{F}_6 \subsetneq \mathfrak{F}_7 \subsetneq \mathfrak{F}_8 \subsetneq \mathfrak{F}_9$. *Each stage strictly expands the frame, and no stage is redundant.*

Proof. Strict expansion at each stage follows from Theorems 3.4–9.4. Non-redundancy: each stage expands in a unique direction.

(i) Stage 3 adds atemporal projections (unique).

(ii) Stage 4 adds presupposition-chain questions—vertical expansion (unique).

(iii) Stage 5 adds anti-questions on restricted spaces—subtractive expansion (unique).

(iv) Stage 6 adds cross-domain imports—horizontal expansion (unique).

(v) Stage 7 adds predictive questions from deductive closure—deductive expansion (unique).

(vi) Stage 8 adds resolution questions from contradictions—dialectical expansion (unique).

(vii) Stage 9 adds separation questions about temporal realisation—reconstructive expansion (unique).

Since each stage produces questions of a type not produced by any other, no stage can be omitted. □

11.1 Worked Example: Four-Element Logical Space

Let $W = \{a, b, c, d\}$ with $|W| = 4$. The partition lattice $\Pi(W)$ has $B_4 = 15$ elements. Consider an initial frame generated by a single partition: $\mathfrak{F}_{\text{init}} = \{\pi_0\}$ where $\pi_0 = \{\{a, b\}, \{c, d\}\}$ (a binary question: "Is the world in $\{a, b\}$ or $\{c, d\}$?").

Stage 2. The question passes the well-formedness test (consistent presuppositions, two answers, sound, open). Proceed.

Stage 3. Suppose a temporal structure τ swaps $a \leftrightarrow b$ (time maps world-a to world-b). The atemporal projection merges a and b: $\pi_0^a = \{\{a, b\}, \{c, d\}\}$, which happens to equal π_0. But the temporal irreducibility question Q_{irred} ("Is time essential?") is new. $|\mathfrak{F}_3| = 2 > 1$.

Stage 4. The presupposition of π_0 is that W has at least two distinguishable classes. The presupposition chain asks: "What makes $\{a, b\}$ and $\{c, d\}$ distinguishable?" This evokes a finer question, e.g., $\pi_1 = \{\{a\}, \{b\}, \{c, d\}\}$. $|\mathfrak{F}_4| \geq 3$.

Stage 5. Remove block $\{a, b\}$ from π_0: the anti-question asks about $W \setminus \{a, b\} = \{c, d\}$, yielding questions on the restricted space. The self-negation question is added. $|\mathfrak{F}_5| \geq 5$.

Stage 6. Suppose a domain $W' = \{x, y, z, w\}$ with isomorphic partition $\pi' = \{\{x, y\}, \{z, w\}\}$. Questions in $\mathcal{N}(\pi')$ not in $\mathfrak{F}_5$ are imported. $|\mathfrak{F}_6| \geq 6$.

Stage 7. Commit to answer $\{a, b\}$ for π_0 and $\{a\}$ for π_1. The deductive closure yields consequences evoking new predictive questions. $|\mathfrak{F}_7| \geq 7$.

Stage 8. If the committed model produces a contradiction (e.g., π_1 says world is a but an imported question implies a is equivalent to b), the resolution question is added. $|\mathfrak{F}_8| \geq 8$.

Stage 9. Reintroduce time. Since τ swaps a and b, the separation question asks what determines whether the system is in a or b at a given time. $|\mathfrak{F}_9| \geq 9$.

Starting from 1 question, the method produces at least 9—a strict expansion at every stage, with each new question of a type unique to its stage. In this example, omitting Stage 6 (isomorphic transfer) leaves the frame without cross-domain questions, which no combination of the other stages can produce since they operate only within the original domain W.

Conjecture (Strong Non-Redundancy). *For every frame $\mathfrak{F}$ satisfying generic conditions, and every stage $k \in \{3, \ldots, 9\}$, there exists a question in $\mathfrak{F}_9$ that is reachable only through stage k—i.e., no combination of the remaining stages, applied in any order, can generate it.*

This conjecture is verified for $|W| = 4$ in the worked example above. A general proof would require showing that the seven expansion operations generate sublattices with pairwise non-trivial symmetric differences, which is an open combinatorial problem.

Why this step matters

The Composition Theorem is the central result of Part I. It establishes that the method is formally grounded with provable expansion properties. Each component is necessary and sufficient for its own type of expansion. The theorem does not claim the method finds the *best* questions—only that it finds questions outside the frame that no proper subset of the method would find. The expansion guarantee is a necessary but not sufficient condition for the method's utility: a method that expands the frame in useless directions would satisfy the theorem. The argument for utility is empirical (Section 12), not formal.

Self-Critique and Resolution

Residual concern. The "generic conditions" may not hold for all phenomena.
Resolution. A phenomenon with no presuppositions, exactly two answers to every question, no temporal structure, and no cross-domain matches would defeat the method. Such a phenomenon would be extraordinary. The conditions are satisfied by the overwhelming majority of phenomena that human inquiry targets.

12 Empirical Demonstration

12.1 Methodology

Each problem was analysed with the RAFASSABRA method prompt (Appendix A) on two independent runs. Eight random words were provided by a human unaware of the problem. The demonstration is *illustrative*, not confirmatory: the sample size (two runs per problem) is insufficient for statistical inference, and the convergence criteria require human judgment. The results are presented as suggestive evidence that the method's structural outputs are more stable than its expressive outputs, and as a proof of concept that the method generates non-trivial questions on substantive problems.

Convergence Criteria

Convergence is assessed at three levels:

(i) **Structural isomorphism of residues.** Two runs converge at Stage 5 if their irreducible residues are formally isomorphic: the same surviving structural features, up to vocabulary.

(ii) **Partition agreement.** Two runs converge at Stage 3 if their atemporal partitions agree on which features are temporal-projective vs. structural.

(iii) **Prediction overlap.** Two runs converge at Stage 7 if their sets of falsifiable predictions have non-empty intersection.

A pair of runs is **structurally convergent** if all three criteria are met, **partially convergent** if two of three are met, and **divergent** if at most one is met.

The Functional Sufficiency Framework

The FSF Kriger (2025) asks, for any phenomenon P: is there a set of functional descriptions $\{f_1, \ldots, f_m\}$ such that $\mathrm{Cn}(\{f_i\}) \supseteq \mathrm{Cn}(\{P\})$? If yes, the phenomenon is a label; if no, it names a distinct entity. This is the formal version of the self-negation question (Stage 5).

12.2 Problem 1: The Nature of Time

Random words: *knot*, *price*, *muscle*, *window*, *dust*, *choir*.

Convergence level: Structural (all three criteria met).

Both runs converged on the residue: time without temporal vocabulary is a partial order on events, structurally isomorphic to logical implication (reflexive, antisymmetric, transitive). Both predicted that the "flow of time" is a temporal projection, not structural.

Divergence occurred in modelling the arrow of time: one run produced an information-theoretic model (past = direction of higher mutual information), the other a boundary-condition model (past = direction of lower entropy). These are complementary, not contradictory.

Knot forced consideration of self-referential temporal structures, exposing the hidden presupposition of linearity. *Window* generated the isomorphism of time as bounded aperture—finite opening with opacity on both sides.

12.3 Problem 2: Consciousness

Random words: *salt*, *envelope*, *gravity*, *carpet*, *bell*, *anchor*.

Convergence level: Structural.

Residue: consciousness = reflexive information structure (fixed point of self-modelling). Both runs applied the FSF independently: "consciousness" survives self-negation only if it names the reflexive structure itself; broader notions (qualia, subjective experience) do not survive.

Divergence: one run treated the hard problem as a limit of formal description; the other as a level-dependent property (like temperature: real at macro level, absent at micro level).

Envelope deformed the ascent: consciousness as boundary separating self from non-self. Without boundary, consciousness collapses into undifferentiated processing. The boundary is essential; the content is not.

12.4 Problem 3: The Nature of Philosophy

Random words: *rust*, *telescope*, *recipe*, *bruise*, *tide*, *seed*.

Convergence level: Structural.

Residue: philosophy is the activity of operating at the boundary of formal coherence—inhabiting undecidable questions in the sense of Gödel. A "solution" to a philosophical problem would move it out of philosophy. Philosophy maintains problems as open questions, preventing premature closure.

Stylistic divergence was maximal (structural convergence with fully divergent expression), confirming the pattern: the method's structural outputs are model-independent; expressive outputs are not.

12.5 Problem 4: Corruption

Random words: *refrigerator*, *aquarium*, *blanket*, *compass*, *volcano*, *thread*.

Convergence level: Structural.

Residue: corruption = resource extraction outside formally sanctioned channels. Structurally isomorphic to thermodynamic entropy. "Eliminating corruption" is formally identical to "eliminating entropy"—a thermodynamic impossibility. Prediction: measures redirecting resource flow succeed; measures attempting elimination fail.

Control experiment. Both models were asked to generate their own "random" words. All 12 self-generated words (*shadow, root, mask, network, mirror, erosion, decay, parasite, facade, web, corrosion, drain*) were semantically related to corruption. None produced structural deformation. By contrast, externally-sourced words produced deformation in 10 of 12 cases across both runs.

This confirms Theorem 10.3: self-generated words have $\|\vec{w}_\perp\| \approx 0$.

Refrigerator forced the question: what does corruption *preserve*? Answer: informal power structures, local equilibria. *Aquarium* generated the transparency paradox: visibility without intervention is observation without effect.

12.6 Problem 5: Dark Matter (Illustrative)

Random words: *wrinkle*, *debt*, *breath*, *lens*, *fossil*, *echo*.

Convergence level: Partial (residue and isomorphism converged; predictions diverged).

Residue: dark matter is a geometric discrepancy between curvature from visible sources and observed curvature. Isomorphism: residual stress in materials science (observed deformation exceeds known forces).

Both runs independently generated the same hypothesis: gravitational effects attributed to dark matter arise from topological features of spacetime (folds, non-trivial topology) rather than unseen matter.

Status: This hypothesis illustrates the method's generative capacity. Its physical evaluation belongs to observational cosmology. The claim is not that the hypothesis is correct but that it is non-trivially generated and independently produced.

12.7 Convergence Analysis

Structural features converge; expressive features diverge universally. This supports the claim that the method's outputs are properties of the method and the problem, not of the specific model.

Feature	Converged	Partial	Diverged
Atemporal residue	5/5	0/5	0/5
Self-negation outcome	4/5	1/5	0/5
Primary isomorphism	4/5	0/5	1/5
Core model	5/5	0/5	0/5
Specific predictions	3/5	1/5	1/5
Expressive features	0/5	0/5	5/5

Table 1: Descriptive counts of convergence across two independent runs on five problems. These are observations, not statistical estimates.

12.8 Problem 6: Love (Independent External Validation)

An independent test of the method was conducted by a third-party model (Gemini) with no involvement from the author. The model was given the paper, asked to analyse "love" using RAFASSABRA, and generated its own random words: *microscope, turbine, mustard, shell, cloud, rust, pendulum, compass.*

Key results by stage:

Stage 3 (atemporal pass) produced: love without time is a topological gluing of two information systems—a geometry of connections where boundaries between self and other become permeable, with identity preserved (forced by the word *mustard*: difference must survive the gluing).

Stage 5 (irreducible residue) produced: after removing sex, friendship, care, shared interests, and habit, what survives is an intentional vector—a directional attention that persists without stimulus. The word *cloud* deformed this into a distributed probability cloud of another system's states perceived as one's own.

Stage 6 (isomorphic transfer) identified quantum entanglement as the structural isomorphism: macroscopic correlation of two information systems, with *rust* generating the question of what produces decoherence in this coupling.

Stage 7 (model) produced: love is a state of coherence between two information systems whose predictive models of the world merge, reducing total computational complexity of survival.

Stage 8 (paradox) identified: if love reduces complexity, why does it often increase it (conflict, drama)? The word *pendulum* resolved this: conflict is calibration—systems testing boundaries of coherence to prevent total absorption.

Final definition: "Love is a time-insensitive topological invariant arising from macroscopic quantum-like entanglement of two information systems, whose function is to overcome the computational boundedness of each system individually through creation of a shared predictive model."

New question generated: "What is the minimum bandwidth of the information channel required to maintain structural entanglement of two persons under conditions of high environmental noise?"

This result is significant for three reasons. (i) It was produced by an executor who had no prior engagement with the method beyond reading the paper—demonstrating that the method is transferable. (ii) The definition is genuinely non-trivial: it translates love from the vocabulary of emotion into information-theoretic and topological language, generating testable predictions (e.g., entangled pairs should show anomalously high information exchange rate and low internal entropy). (iii) The final question—about minimum channel bandwidth—is a concrete, formalizable research question that could not have been asked within the standard psychological or biological frames for love. This is precisely the kind of frame-breaking question the method is designed to produce.

12.9 RAFASSABRA as a De-Templating Tool for Artificial Intelligence

The RAFASSABRA method was designed for human inquiry, but its formal structure addresses a problem that is arguably more severe in artificial intelligence: the inability to exit trained cognitive frames.

The template problem. Large language models (LLMs) are trained on vast corpora of human text. This training produces powerful pattern-matching, but it also produces powerful pattern-*repetition*. When asked to analyse a phenomenon, an LLM will retrieve the most statistically likely framing from its training data—which is, by definition, the most conventional one. The model does not lack knowledge; it lacks the capacity to reorganise what it knows into non-standard configurations. Its cognitive frame is the statistical average of all frames it has seen.

This is precisely the condition the RAFASSABRA method is designed to break. Each stage targets a specific dimension of template thinking:

Stage 1 (Random Words) injects external perturbation that is statistically independent of the model's training distribution—exactly the kind of input the model cannot generate for itself (as the control experiment in Section 12 demonstrated: self-generated "random" words cluster within the active frame).

Stage 2 (Structural Attack) forces the model to question its own presuppositions, counteracting the tendency to accept the user's framing uncritically.

Stage 3 (Atemporal Pass) removes temporal and causal language, which is the dominant organisational template in training data (narrative structure). This forces the model into a non-narrative mode of analysis.

Stage 5 (Irreducible Residue) strips away the features that make a phenomenon recognisable to a pattern-matcher, leaving only what is structurally necessary—the opposite of template retrieval.

Stage 6 (Isomorphic Transfer) forces cross-domain structural mapping, which requires the model to use its knowledge base in a combinatorial rather than retrieval mode.

The empirical evidence supports this interpretation. Problem 6 (Love), analysed by an independent AI model using RAFASSABRA, produced a definition ("time-insensitive topological invariant arising from macroscopic quantum-like entanglement") that no standard prompt would elicit. The model's training data contains millions of texts about love; none frames it in information-theoretic topology. The method forced the model out of its statistical attractor basin.

The broader implication. If AI systems are to contribute to scientific discovery rather than merely summarise existing knowledge, they need methods for systematic frame-breaking. RAFASSABRA provides one such method, with formal guarantees. The random word mechanism is particularly important for AI: a human can have a surprising experience, travel to an unfamiliar place, or talk to someone with a radically different worldview. An LLM, confined to its training distribution, cannot. The external random word is the AI's equivalent of an unexpected encounter—a forced collision with something outside its frame.

This suggests a practical application: RAFASSABRA can be used as a *prompt architecture* for AI-assisted research, not by asking the model "be creative," but by structurally constraining it through nine stages that make template-retrieval impossible and frame-breaking inevitable.

12.10 Limitations

(i) The control experiment (self-generated words) was conducted for one problem only. Future work should replicate across all five.

(ii) Two models is a minimum for convergence analysis. Additional models would strengthen the results.

(iii) The convergence criteria require human judgement. Inter-rater reliability should be assessed.

(iv) The dark matter hypothesis has not been evaluated against observational data.

Part II
The Priority of Questions over Answers

13 Introduction: The Forgotten Priority

Every formal system begins with axioms. This is so deeply embedded in mathematical and scientific practice that it appears self-evident: one states what is assumed to be true, then derives consequences. Yet there is something that precedes every axiom—the question that motivated it. Euclid did not begin with five postulates; he began with the question of what must be assumed for geometric reasoning to proceed. Newton did not begin with three laws; he began with the question of why celestial and terrestrial motions follow the same patterns. Gödel did not begin with incompleteness; he began with the question of whether a formal system can prove its own consistency.

In each case, the question shaped the axioms that followed—determined what would count as a starting point, what would count as a valid derivation, and what would count as a result. The axioms were answers to questions that were never made explicit, and whose structure was never examined.

This Part investigates a thesis that, despite its simplicity, has significant consequences for logic, epistemology, and the methodology of inquiry: the space of formulable questions constrains the space of possible truths more fundamentally than axiomatic declaration.

13.1 Three Claims

The paper advances three claims, each building on the previous:

Claim A (Priority). Questions are structurally prior to answers. We define *structural priority* as follows: X is structurally prior to Y if (i) the functor from the category of X-structures to Y-structures is not faithful, (ii) the relational structure of X (lattice ordering, directed graph, Lie algebra) has no counterpart in Y, and (iii) Y cannot be reconstructed from X without presupposing X-type concepts. Under four independent formalisations and three mathematical frameworks, questions satisfy all three conditions relative to answers. Whether structural priority entails ontological priority in a metaphysical sense is a philosophical question we flag as open; the formal content of Claim A is the structural asymmetry.

Claim B (Constraint). Cognitive frames restrict the space of formulable questions more severely than they restrict the space of available answers. An agent operating within a frame cannot formulate questions outside that frame, even when the answers to such questions are derivable from information already available.

Claim C (Method). There exists a systematic method for expanding the space of formulable questions beyond the current frame. The method—the RAFASSABRA method—provably expands the question space.

Note on logical dependence. The three claims form a coherent narrative—A explains *why* the bottleneck is interrogative, B explains *where* it bites hardest, C addresses it—but the chain is motivational, not deductive. A practitioner need not accept structural priority to benefit from systematic question expansion. The method stands on its own empirical and practical merits.

13.2 Intellectual Context

The priority of questions over answers has been explored in erotetic logic Belnap and Steel (1976); Hintikka (1999); Wisniewski (1995), hermeneutics Gadamer (1960); Collingwood (1939), quantum mechanics von Neumann (1932), cognitive science Kahneman (2011); Nickerson (1998); Carey (2009), and the tree-top meta-method Kriger (2026). The practical technique of forcing alien associations to break fixed patterns has antecedents in De Bono's lateral thinking, the Geneplore model of creative cognition, and the TRIZ method's analogical transfer. The RAFASSABRA method's distinctive contribution is the *formal guarantee* of frame-externality (Theorem 10.3), which these earlier methods lack. No prior work has produced a unified formal treatment linking erotetic priority to frame restriction to a provably expanding inquiry method. This paper provides one.

14 Four Formalisations of "Question"

We adopt four independent formalisations of the concept "question." Each captures a different aspect of interrogative structure. If the priority thesis holds under all four, it is robust against definitional choice.

14.1 Formalisation 1: Partition Semantics

Definition 14.1 (Question as Partition)**.** Let W be a non-empty set of possible worlds. A **question** Q is a partition $\pi_Q = \{B_1, B_2, \ldots, B_k\}$ of W: a family of non-empty, pairwise disjoint subsets whose union is W. Each block B_i is a **complete answer**: it specifies exactly which worlds are compatible with that answer.

Definition 14.2 (Refinement)**.** Partition π_1 **refines** π_2 (written $\pi_1 \preceq \pi_2$) if every block of π_1 is contained in some block of π_2. Refinement means: answering π_1 entails answering π_2.

The partition lattice $(\Pi(W), \preceq)$ is a complete lattice. Its minimum is the discrete partition $\Delta = \{\{w\} : w \in W\}$ (the most specific question: "Which exact world are we in?") and its maximum is the indiscrete partition $\nabla = \{W\}$ (the trivial question with only one answer). The lattice operations are: meet $\pi_1 \wedge \pi_2$ (asking both questions simultaneously) and join $\pi_1 \vee \pi_2$ (the weakest common question).

This formalisation originates with Hintikka (1976) and was developed by Groenendijk and Stokhof (1997).

14.2 Formalisation 2: Erotetic Implication (Inferential Erotetic Logic)

Definition 14.3 (Question as Answer Set)**.** A **question** Q in a declarative language $\mathcal{L}_d$ is identified with its set of **direct answers** $\mathrm{dQ} = \{\alpha_1, \ldots, \alpha_k\} \subseteq \mathcal{L}_d$, where each α_i is a sentence that completely resolves Q.

Definition 14.4 (Presupposition and Evocation)**.** The **presupposition** of Q is $\mathrm{presup}(Q) = \bigvee_{i=1}^{k} \alpha_i$ (the disjunction of all direct answers). A set of sentences X **evokes** Q (written $\mathrm{Ev}(X, Q)$) if and only if:

(i) $X \vdash \bigvee_i \alpha_i$ (soundness: the presupposition follows from X), and

(ii) $X \nvdash \alpha_i$ for each individual i (openness: no single answer is settled by X).

Definition 14.5 (Erotetic Implication)**.** Question Q_1 **implies** Q_2 relative to X (written $\mathrm{Im}(Q_1, X, Q_2)$) if every direct answer to Q_1, together with X, either answers Q_2 or evokes Q_2 as a further question. This induces a directed graph on questions—the strategic structure of inquiry.

This formalisation follows Wisniewski (1995); Wiśniewski (2013) and Belnap and Steel (1976).

14.3 Formalisation 3: Interrogative Model of Inquiry

Definition 14.6 (Interrogative Game)**.** An **interrogative game** $G = (\mathcal{L}, T, \Omega, \mathcal{Q}, \mathcal{R})$ consists of: a first-order language $\mathcal{L}$; an initial theory $T \subseteq \mathcal{L}$; an oracle Ω (nature, an experiment, an informant); an interrogative capacity $\mathcal{Q}$ (the set of questions the agent can ask Ω); and derivation rules $\mathcal{R}$.

The agent alternates between **deductive moves** (deriving consequences from current knowledge using $\mathcal{R}$) and **interrogative moves** (asking Ω a question $Q \in \mathcal{Q}$ and receiving an answer). The interrogative closure is:

$$\mathrm{Cn}_q(T, \mathcal{Q}) = \mathrm{Cn}_d\big(T \cup \{\alpha : \exists Q \in \mathcal{Q},\ \Omega \text{ answers } \alpha \text{ to } Q\}\big).$$

The key insight: $\mathrm{Cn}_q(T, \mathcal{Q})$ depends on $\mathcal{Q}$. If $\mathcal{Q}_1 \subsetneq \mathcal{Q}_2$, then generically $\mathrm{Cn}_q(T, \mathcal{Q}_1) \subsetneq \mathrm{Cn}_q(T, \mathcal{Q}_2)$. Deductive power cannot compensate for interrogative poverty.

This formalisation follows Hintikka (1999).

14.4 Formalisation 4: Questions as Operators

Definition 14.7 (Question as Observable)**.** Let $\mathcal{H}$ be a finite-dimensional Hilbert space. A **question** is a self-adjoint operator $\hat{Q} : \mathcal{H} \to \mathcal{H}$. Its spectral decomposition $\hat{Q} = \sum_{i=1}^{k} \lambda_i P_i$ yields: **possible answers** (eigenvalues λ_i) and **answer contexts** (eigenprojections P_i). The probability of answer λ_i given state ρ is $p_i = \mathrm{tr}(\rho P_i)$.

Definition 14.8 (Non-Commutativity)**.** Questions $\hat{Q}_1, \hat{Q}_2$ are **compatible** if $[\hat{Q}_1, \hat{Q}_2] = 0$ (they can be asked simultaneously without interference) and **incompatible** if $[\hat{Q}_1, \hat{Q}_2] \neq 0$ (asking one disturbs the answers to the other). Incompatibility imposes the generalised uncertainty relation Maassen and Uffink (1988).

This formalisation extends von Neumann (1932) and has been applied to cognitive science by Busemeyer and Bruza (2012). Its use here is as a *mathematical* model of non-commutative questions, not as a physical claim about quantum cognition.

14.5 Relations Among Formalisations

The four formalisations are not independent descriptions of the same object; they are complementary views highlighting different features:

	F1: Partition	**F2: Erotetic**	**F3: Interrogative**	**F4: Operator**
Question	Partition of W	Answer set dQ	Move in game	Self-adjoint op.
Answer	Block B_i	Sentence α_i	Oracle reply	Eigenvalue λ_i
Structure	Lattice	Directed graph	Game tree	Lie algebra
Key feature	Refinement	Evocation	Capacity bound	Non-commutativity
Priority signature	Lattice vs. non-lattice	Directed graph vs. preorder	Capacity-dependent closure	Non-commutative algebra vs. scalar set

Table 2: Summary of the four formalisations.

The priority thesis is proved under each formalisation independently. The convergence of results across formalisations (demonstrated in Sections 15–17) establishes that priority is a structural fact, not an artefact of definition.

14.6 Limitation: Selection of Formalisations

All four formalisations come from traditions that treat questions as primitive or structurally central. Proposition-first frameworks—standard Bayesian epistemology (where priors and likelihoods are foundational), classical model theory (where satisfaction relations are primary), propositional dynamic logic—are not represented. Our claim of robustness is therefore robustness within the erotetic family. We conjecture that the structural asymmetry survives under proposition-first frameworks, because even there the question “which proposition?” must precede the selection of any specific proposition, reintroducing interrogative structure at the meta-level. Testing this conjecture under a formally constructed proposition-first framework is the most important direction for future work.

15 Priority of Questions: Set-Theoretic and Topological Proofs

We now prove the priority thesis under each of the four formalisations, using set-theoretic and topological methods. For each, we state and prove a theorem, then attempt a counter-proof and analyse its failure.

15.1 Under Partition Semantics (F1)

Theorem 15.1 (Partition Priority)**.** *Let W be a set with $|W| \geq 2$. Let $\Pi(W)$ denote the set of all partitions of W, and let $\mathcal{A}(W) = \bigcup_{\pi \in \Pi(W)} \pi$ denote the set of all blocks appearing in any partition. Then $\Pi(W)$ has a complete lattice structure that $\mathcal{A}(W)$ lacks.*

Lemma 15.2 (Cardinality of Partition and Block Spaces)**.** *For finite W with $|W| = n$: (i) $|\Pi(W)| = B_n$ (the Bell number); (ii) $|\mathcal{A}(W)| = 2^n - 1$; (iii) B_n grows super-exponentially while $2^n - 1$ grows merely exponentially, so $B_n > 2^n - 1$ for all $n \geq 15$.*

Proof. Part (i): B_n counts partitions of an n-element set. Part (ii): every non-empty subset of W can serve as a block in some partition (pair it with its complement, then partition the complement arbitrarily), so the set of all possible blocks is exactly $\mathcal{P}(W) \setminus \{\emptyset\}$. Part (iii): by the de Bruijn asymptotic, $B_n \sim (n/\ln n)^n$; the crossover occurs at $n = 15$ where $B_{15} =$ 1,382,958,545 and $2^{15} - 1 =$ 32,767. □

The cardinality comparison is informative but insufficient for priority in the structural sense. The deeper result concerns lattice structure.

Theorem 15.3 (Lattice Structure of the Question Space)**.** *$(\Pi(W), \preceq)$ forms a complete lattice. $\mathcal{A}(W)$ does not form a lattice under any natural ordering. The lattice structure encodes inferential relationships among questions that have no counterpart in the answer space.*

Proof. That $(\Pi(W), \preceq)$ is a complete lattice is classical. The minimum is the discrete partition $\{\{w\} : w \in W\}$; the maximum is $\{W\}$. For any two partitions π_1, π_2, the meet $\pi_1 \wedge \pi_2$ has blocks that are all non-empty intersections $B_i \cap C_j$, and the join $\pi_1 \vee \pi_2$ is the finest partition coarser than both. Completeness follows because intersections of equivalence relations are equivalence relations.

Now consider $\mathcal{A}(W)$. Under set inclusion $\subseteq$, the non-empty subsets form a join-semilattice (unions exist) but not a lattice: intersections of two blocks from different partitions may be empty, which is not in $\mathcal{A}(W)$. Therefore $\Pi(W)$ possesses a complete lattice structure that $\mathcal{A}(W)$ lacks. □

Why this step matters

The lattice structure is precisely the structure of inferential relationships among questions: $\pi_1 \preceq \pi_2$ means "answering π_1 entails answering π_2." The meet and join correspond to asking both questions simultaneously and asking the weakest common question, respectively. There is no analogous structure on individual answers.

> ***Counter-Argument.*** *Every partition can be represented as a set of blocks. Therefore $\Pi(W) \subseteq \mathcal{P}(\mathcal{A}(W))$: every question is a set of answers, and the answer space generates the question space.*

The counter-argument is formally correct but conflates representability with reducibility. (i) Not every subset of $\mathcal{A}(W)$ is a partition—the disjointness and covering constraints are properties of the question space. (ii) The lattice structure of $\Pi(W)$ is not present in $\mathcal{A}(W)$ or $\mathcal{P}(\mathcal{A}(W))$. (iii) A partition includes the requirement that blocks are disjoint and covering—a structural requirement that transcends the individual blocks, as a symphony transcends its notes.

15.2 Under Erotetic Implication (F2)

Theorem 15.4 (Erotetic Priority). *Let $\mathcal{L}_d$ be a countable first-order language. For each consistent theory X, let $\mathrm{Ev}(X)$ be the set of evoked questions and $\mathrm{Cn}(X)$ the deductive closure. Then: (i) $\mathrm{Ev}(X)$ collectively requires information not in $\mathrm{Cn}(X)$; (ii) erotetic implication induces a directed graph on questions with no counterpart on declarative sentences.*

Proof. For (i): Let $Q \in \mathrm{Ev}(X)$ with direct answers $\{\alpha_1, \ldots, \alpha_k\}$. By evocation, $\bigvee_i \alpha_i \in \mathrm{Cn}(X)$ but $\alpha_i \notin \mathrm{Cn}(X)$ for each i. Each Q represents a k-fold branching whose resolution requires information outside $\mathrm{Cn}(X)$. The question space maps the boundary of ignorance; the answer space maps the interior of knowledge.

For (ii): Erotetic implication $\mathrm{Im}(Q_1, X, Q_2)$ induces a directed graph $G = (\mathrm{Ev}(X), E)$ encoding the strategic structure of inquiry. Entailment on $\mathrm{Cn}(X)$ is a preorder, not a dynamic graph where traversing an edge (answering a question) changes the vertex set (new questions evoked). □

> ***Counter-Argument.*** *Questions in IEL are defined by their direct answers $\mathrm{dQ} \subseteq \mathcal{L}_d$. The question space is a subset of $\mathcal{P}(\mathcal{L}_d)$, so answers are prior.*

The counter-argument correctly identifies definitional dependence but fails structurally: (i) not every subset of $\mathcal{L}_d$ is a well-formed question—identifying which subsets qualify requires erotetic concepts; (ii) the evocation relation is not derivable from entailment alone; (iii) the directed graph of erotetic implication has no source in $\mathcal{L}_d$.

15.3 Under the Interrogative Model of Inquiry (F3)

Theorem 15.5 (Interrogative Priority). *Let $G = (\mathcal{L}, T, \Omega, \mathcal{R})$ be an interrogative game. Then: (i) $\mathrm{Cn}_d(T) \subseteq \mathrm{Cn}_q(T, \mathcal{Q})$; (ii) for $\mathcal{Q}_1 \subsetneq \mathcal{Q}_2$, $\mathrm{Cn}_q(T, \mathcal{Q}_1) \subseteq \mathrm{Cn}_q(T, \mathcal{Q}_2)$; (iii) there exist T, ϕ such that $\phi \in \mathrm{Cn}_q(T, \mathcal{Q}_1 \cup \{Q\})$ but $\phi \notin \mathrm{Cn}_q(T, \mathcal{Q}_1)$ regardless of deductive resources.*

Proof. For (iii): Let $T = \{P(a), \forall x(P(x) \wedge R(x) \to P(f(x)))\}$ and $\phi = P(f(a))$. Deriving ϕ requires $R(a)$, which is not in $\mathrm{Cn}_d(T)$. The only way to obtain $R(a)$ is to ask Q_R: "Does $R(a)$ hold?" If $Q_R \notin \mathcal{Q}_1$, then ϕ is unreachable. No deductive power compensates. □

> ***Counter-Argument.*** *If the oracle's answers were added to T from the start (forming $T^* = T \cup \{\alpha : \Omega$ would answer $\alpha\}$), then $\mathrm{Cn}_d(T^*) = \mathrm{Cn}_q(T, \mathcal{Q})$ and questions add nothing.*

Failure: (i) Constructing T^* requires knowing answers, which requires asking questions—circular. (ii) Even given T^*, relevance assessment is interrogative (the frame problem). (iii) By Gödel's theorem Gödel (1931), no T^* is deductively complete.

15.4 Under the Operator Formalism (F4)

Theorem 15.6 (Operator Priority)**.** *Let $\mathcal{H}$ have $\dim(\mathcal{H}) = n \geq 2$. Then: (i) the question algebra $\mathcal{Q}(\mathcal{H})$ is n^2-dimensional while the answer set $\Lambda^* = \mathbb{R}$; (ii) distinct operators can have identical eigenvalue sets but different eigenbases (e.g., σ_z and σ_x both have eigenvalues $\{+1, -1\}$); (iii) non-commuting questions impose the uncertainty relation $\Delta Q_1 \cdot \Delta Q_2 \geq \frac{1}{2}|\langle[\hat{Q}_1, \hat{Q}_2]\rangle|$ Maassen and Uffink (1988).*

Proof. (i) The $n \times n$ Hermitian matrices form an n^2-dimensional real vector space. For $n \geq 2$, $n^2 > 1 = \dim(\mathbb{R})$. (ii) $\sigma_z = \mathrm{diag}(1, -1)$ and $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ have the same spectrum but orthogonal eigenbases. (iii) $[\sigma_z, \sigma_x] = 2i\sigma_y \neq 0$, so the uncertainty relation is non-trivial. □

> ***Counter-Argument.*** *Every self-adjoint operator is determined by its spectral decomposition $\hat{Q} = \sum_i \lambda_i P_i$. Questions reduce to projections and eigenvalues.*

Failure: (i) The choice of decomposition is not determined by the state space. (ii) The projections P_i depend on $\hat{Q}$—one cannot assemble operators from projections without knowing which go together. (iii) The commutator is a relational property not deducible from individual spectra.

15.5 Set-Theoretic Convergence

Proposition 15.7 (Set-Theoretic Convergence)**.** *Under all four formalisations: (i) the question space contains information the answer space does not (lattice structure, erotetic graph, interrogative capacity, non-commutative algebra); (ii) distinct questions can have identical answer sets; (iii) the question space cannot be reconstructed from the answer space without presupposing interrogative concepts.*

16 Priority of Questions: Category-Theoretic Proofs

We reprove the priority thesis using category theory. This framework reveals structural relationships invisible to set theory: questions and answers are objects in different categories connected by functors that are structurally asymmetric.

16.1 Categorical Preliminaries

A **category** $\mathbf{C}$ consists of objects, morphisms, identity morphisms, and a composition law satisfying associativity and identity. A **functor** $F : \mathbf{C} \to \mathbf{D}$ preserves identity and composition. A functor is *faithful* if injective on morphism sets and *full* if surjective on morphism sets. A **natural transformation** $\eta : F \Rightarrow G$ assigns to each object A a morphism $\eta_A : F(A) \to G(A)$ respecting composition.

16.2 The Categories of Questions and Answers

Definition 16.1 (Category of Answers $\mathbf{Ans}(W)$)**.** Objects: non-empty subsets $A \subseteq W$ (propositions). Morphisms: inclusions $A \hookrightarrow B$ whenever $A \subseteq B$. This is the poset category $(\mathcal{P}(W) \setminus \{\emptyset\}, \subseteq)$.

Definition 16.2 (Category of Questions $\mathbf{Qu}(W)$)**.** Objects: partitions π of W. Morphisms: $\pi_1 \to \pi_2$ whenever π_1 refines π_2. This is the partition lattice as a category.

Definition 16.3 (Answer Functor $\mathcal{A} : \mathbf{Qu}(W) \to \mathbf{Set}$)**.** On objects: $\mathcal{A}(\pi) = \pi$ (the set of blocks). On morphisms: if $\pi_1 \preceq \pi_2$, then $\mathcal{A}(\pi_1 \to \pi_2)$ is the surjection sending each block $B \in \pi_1$ to the unique $C \in \pi_2$ with $B \subseteq C$.

16.3 Under Partition Semantics (F1)

Theorem 16.4 (Categorical Partition Priority)**.** *The answer functor $\mathcal{A}$ is not faithful and does not reflect isomorphisms.*

Proof. Non-faithfulness. Let $|W| = 4$, $W = \{a, b, c, d\}$. Let $\pi_1 = \{\{a\}, \{b\}, \{c\}, \{d\}\}$, $\pi_2 = \{\{a, b\}, \{c, d\}\}$, $\pi_3 = \{\{a, c\}, \{b, d\}\}$. Both $\pi_1 \preceq \pi_2$ and $\pi_1 \preceq \pi_3$. Under $\mathcal{A}$, both morphisms become surjections $4 \twoheadrightarrow 2$—the same abstract type. Yet f and g are distinct morphisms in $\mathbf{Qu}(W)$.

Non-reflection. $\mathcal{A}(\pi_2) = \{\{a, b\}, \{c, d\}\}$ and $\mathcal{A}(\pi_3) = \{\{a, c\}, \{b, d\}\}$ are both 2-element sets, hence isomorphic in $\mathbf{Set}$. But π_2 and π_3 are not isomorphic in $\mathbf{Qu}(W)$ (neither refines the other). □

> ***Counter-Argument.*** *There exists a functor $\mathcal{B} : \mathbf{Ans}(W) \to \mathbf{Qu}(W)$ sending each proposition A to the yes/no question $\pi_A = \{A, W \setminus A\}$. This is faithful and injective on objects.*

The embedding exists but its image consists only of binary partitions—a small sublattice. It maps *into* $\mathbf{Qu}(W)$, meaning answers acquire meaning only when interpreted as responses to questions. The composite $\mathcal{A} \circ \mathcal{B}$ collapses all propositions to 2-element sets. The round-trip loses structure. This asymmetry of composability is a categorical signature of priority.

16.4 Under Erotetic Implication (F2)

Theorem 16.5 (Categorical Erotetic Priority)**.** *The erotetic category $\mathbf{Ero}(X)$ (objects: evoked questions; morphisms: erotetic implications) has: (i) multiple distinct paths between questions; (ii) cycles (a question may re-evoke itself through implication chains); (iii) neither property holds in $\mathbf{Dec}(X)$ (the declarative category under entailment, which is a preorder).*

Proof. (i) If Q_1's first answer resolves Q_3 directly while Q_1's second answer evokes Q_2 which implies Q_3, there are two distinct paths $Q_1 \to Q_3$ corresponding to different inquiry strategies. (ii) Answering Q_1 may evoke Q_2; answering Q_2 may evoke Q_1 in the expanded theory $X \cup \{\alpha_j\}$. (iii) In $\mathbf{Dec}(X)$, cycles collapse to equivalence classes; multiple paths reduce to single morphisms. □

16.5 Under the Interrogative Model (F3)

Theorem 16.6 (Categorical Interrogative Priority)**.** *The inquiry category* $\mathbf{Inq}(G)$ *is not equivalent to its deductive subcategory* $\mathbf{Ded}(G)$*: (i)* $\mathbf{Inq}(G)$ *has objects not in* $\mathbf{Ded}(G)$*; (ii) the inclusion* $\iota : \mathbf{Ded}(G) \hookrightarrow \mathbf{Inq}(G)$ *has no left adjoint; (iii)* $\mathbf{Inq}(G)$ *is generically more connected.*

Proof. (i) By Theorem 15.5(iii), theory states reachable only through interrogative moves exist in $\mathbf{Inq}(G) \setminus \mathbf{Ded}(G)$. (ii) A left adjoint would provide a "best deductive approximation" for every interrogative state; this fails when answers are not deductively reachable. (iii) Interrogative moves create edges between deductively disconnected states. □

16.6 Under the Operator Formalism (F4)

Theorem 16.7 (Categorical Operator Priority)**.** *The spectrum functor* $\Sigma : \mathbf{Obs}(\mathcal{H}) \to \mathbf{Set}$ *(sending each observable to its eigenvalue set) is not faithful, does not reflect isomorphisms, and the commutator structure of* $\mathbf{Obs}(\mathcal{H})$ *has no image under* Σ*.*

Proof. Non-faithfulness and non-reflection follow from Theorem 15.6(ii): σ_z and σ_x have the same spectrum but are distinct non-isomorphic objects. The commutator $[\hat{Q}_1, \hat{Q}_2]$ is a bilinear operation encoding incompatibility; spectra have no notion of commutativity. The entire Lie algebra of observables is invisible at the answer level. □

16.7 Category-Theoretic Convergence

Proposition 16.8 (Category-Theoretic Convergence)**.** *Under all four formalisations, the functor from questions to answers exhibits: (i) non-faithfulness; (ii) non-reflection of isomorphisms; (iii) loss of relational structure (lattice ordering, directed graph, inquiry connectivity, Lie algebra). These are categorical invariants independent of formalisation.*

17 Priority of Questions: Information-Theoretic Proofs

We reprove the priority thesis using information theory Shannon (1948). This framework is quantitative: instead of proving that question spaces have richer structure, we prove that questions carry more information than answers.

17.1 Preliminaries

The Shannon entropy of a discrete random variable X is $H(X) = -\sum_x p(x) \log_2 p(x)$. The mutual information between X and Y is $I(X;Y) = H(X) - H(X|Y)$. A question Q with partition $\pi_Q = \{B_1, \ldots, B_k\}$ and state distribution μ has information content $H(Q) = -\sum_i \mu(B_i) \log_2 \mu(B_i)$.

17.2 Under Partition Semantics (F1)

Theorem 17.1 (Information-Theoretic Partition Priority)**.** *For $|W| = n \geq 2$ with distribution μ: (i) the question entropy space $\{H(Q) : Q \in \Pi(W)\}$ is the interval $[0, \log_2 n]$; (ii) the expected information from answering Q is exactly $H(Q)$; (iii) specifying a question (the partition π_Q) requires strictly more information than specifying any of its answers (a block B_i).*

Proof. (i) $H(Q) = 0$ for the trivial partition; $H(Q) = \log_2 n$ for the discrete partition with uniform μ. Intermediate values are achieved by splitting/merging blocks. (ii) $\mathbb{E}[I(B_i)] = \sum_i \mu(B_i) \log_2(1/\mu(B_i)) = H(Q)$. (iii) The Kolmogorov complexity of π_Q satisfies $K(\pi_Q) \geq \max_i K(B_i) + \Omega(1)$ since π_Q contains multiple blocks plus the disjointness/covering constraint. □

> ***Counter-Argument.*** *The answer updates the state distribution; the question merely frames the inquiry. Questions are scaffolding, not substance.*

Failure: (i) The "framing" determines which dimension of uncertainty to reduce—two questions with the same entropy may address entirely disjoint dimensions. (ii) In information geometry, different questions correspond to different projections of μ; a projection direction contains more information about the space's structure than any projected value.

17.3 Under Erotetic Implication (F2)

Theorem 17.2 (Information-Theoretic Erotetic Priority)**.** *For consistent theory X with $|\mathrm{Ev}(X)| \geq 2$: (i) each evoked question has $H(Q|X) > 0$; (ii) the total interrogative entropy $H_{\mathrm{ero}}(X) = \sum_{Q \in \mathrm{Ev}(X)} H(Q|X)$ strictly exceeds $H(X)$; (iii) the mutual information matrix $\{I(Q_i; Q_j|X)\}$ is a property of the question space with no answer-space counterpart.*

Proof. (i) By evocation, no α_i is entailed, so μ assigns positive probability to models satisfying different α_i's, giving $H(Q|X) > 0$. (ii) $H_{\mathrm{ero}}(X) \geq H(X)$ with equality only when all evoked questions partition identically (contradicting distinctness). (iii) $I(Q_i; Q_j|X) = H(Q_i|X) + H(Q_j|X) - H(Q_i, Q_j|X) \neq 0$ when questions address overlapping aspects of knowledge. □

17.4 Under the Interrogative Model (F3)

Theorem 17.3 (Information-Theoretic Interrogative Priority)**.** *The interrogative information gain $\Delta H(\mathcal{Q}) = H_d(T) - H_q(T, \mathcal{Q})$ is monotonically non-decreasing in $|\mathcal{Q}|$. There exists a critical $|\mathcal{Q}^*|$ (the interrogative complexity of ϕ relative to T) below which ϕ cannot be determined with certainty.*

Proof. Each interrogative move yields an answer updating μ with non-negative information gain. Enlarging $\mathcal{Q}$ adds moves that either reduce entropy further or leave it unchanged. The minimum number of binary questions needed to distinguish ϕ-models from $\neg\phi$-models is $\lceil \log_2 m \rceil$ where m is the number of equivalence classes. □

Counter-Argument. *Bayesian updating requires no interrogative structure; the prior and likelihood function suffice.*

Failure: (i) The likelihood function presupposes the question "Which hypothesis is correct?" (ii) Choosing which experiment to perform is choosing a question. (iii) Bayesian updating tells you how to update given evidence; it does not tell you what to ask.

17.5 Under the Operator Formalism (F4)

Theorem 17.4 (Information-Theoretic Operator Priority)**.** *For* $\dim(\mathcal{H}) = n$*: (i) specifying a question requires* $\Theta(n^2)$ *parameters vs.* n *for the eigenvalue set; (ii) the Maassen–Uffink bound* $H(\hat{Q}_1, \rho) + H(\hat{Q}_2, \rho) \geq -2\log_2 c$ *depends on eigenbases, not eigenvalues Maassen and Uffink (1988); (iii) the question algebra* $\mathfrak{su}(n)$ *has dimension* $n^2 - 1$*, requiring* $O(n)$ *measurements for state tomography.*

Proof. (i) An $n \times n$ Hermitian matrix has n^2 real parameters; its eigenvalue set has n. The ratio is n, growing linearly. (ii) The constant $c = \max_{i,j} |\langle \phi_i | \chi_j \rangle|$ is geometric—a property of question-pairs. For mutually unbiased bases, $c = 1/\sqrt{n}$ and the bound is $\log_2 n$ bits. (iii) Complete tomography requires $n+1$ mutually unbiased bases—a specific set of questions whose choice is determined by the question algebra, not by answers. □

17.6 Information-Theoretic Convergence

Proposition 17.5 (Information-Theoretic Convergence)**.** *Under all four formalisations: (i)* ***complexity asymmetry****: the question space has higher Kolmogorov complexity (or parametric dimension); (ii)* ***directional asymmetry****: questions determine the direction of information gain, answers the magnitude; (iii)* ***relational information****: the question space carries mutual information structure, entropic uncertainty relations, and correlation geometry absent from the answer space.*

These converge with the set-theoretic (Proposition 15.7) and category-theoretic (Proposition 16.8) results:

Property	**Set-Theoretic**	**Category-Theoretic**	**Info-Theoretic**
$Q > A$	Lattice vs. poset	Non-faithful functor	Complexity asymmetry
Distinct Q, same A	Incomparable π	Non-reflected isos	Direction vs. magnitude
Relational structure	Refinement ordering	Lie algebra / paths	Mutual information

Table 3: Convergence across three mathematical frameworks.

Part III

Cognitive Frames as Interrogative Constraints

18 Formalisation of Cognitive Frames

A cognitive frame is a restriction on the space of formulable questions. This Part proves that such restrictions affect questions more severely than answers, that the restriction is a mathematical necessity for bounded agents, and that the asymmetry holds across all three mathematical frameworks.

18.1 Set-Theoretic Formalisation

Definition 18.1 (Cognitive Frame)**.** A **cognitive frame** on logical space W is a sublattice $\mathfrak{F} \subseteq \Pi(W)$: a non-empty set of partitions closed under meet $\wedge$ and join $\vee$. The frame is *generated* by a set $G = \{\pi_1, \dots, \pi_r\}$ if $\mathfrak{F} = \langle G \rangle$ (the smallest sublattice containing G).

Definition 18.2 (Frame-Accessible Answers)**.** The **frame-accessible answer set** is $\mathcal{A}(\mathfrak{F}) = \bigcup_{\pi \in \mathfrak{F}} \pi$: the set of all blocks appearing in any partition in the frame.

Definition 18.3 (Restriction Ratios)**.** For frame $\mathfrak{F}$: the **question restriction ratio** is $r_Q = 1 - |\mathfrak{F}|/|\Pi(W)|$ and the **answer restriction ratio** is $r_A = 1 - |\mathcal{A}(\mathfrak{F})|/|\mathcal{A}(\Pi(W))|$.

Theorem 18.4 (Frame Restriction Asymmetry, Small n)**.** *For $|W| = n$ with $4 \leq n \leq 7$ and frames of intermediate richness ($1 < |\mathfrak{F}| < |\Pi(W)|$), generically $r_Q > r_A$: frames exclude a larger fraction of questions than of answers. Here "generically" means for all but a measure-zero set of frames under uniform random sampling of generators from $\Pi(W)$. This is verified computationally across 500 random trials per (n, r) pair.*

We conjecture that the result holds for all $n \geq 4$:

Conjecture (Frame Restriction Asymmetry, General n). *For all $|W| = n \geq 4$ and frames of intermediate richness, generically $r_Q > r_A$. Moreover, the gap $r_Q - r_A$ increases with n.*

The conjecture is supported by the asymptotic argument below and by the computational evidence, but a complete analytical proof for arbitrary n remains an open problem.

Proof. Let $\mathfrak{F} = \langle G \rangle$ with $|G| = r$ generators. Each generator π_i has k_i blocks. The answer set includes: (a) all blocks of all generators ($\sum_i k_i$ blocks, with possible overlap); (b) all non-empty intersections of blocks from different generators (up to $\prod_i k_i$ new blocks from cross-intersections); (c) all complements of such intersections that appear as blocks in the lattice closure.

The partition set includes: (a) the r generators; (b) all pairwise meets and joins; (c) the lattice closure.

The key asymmetry: block-generation is *multiplicative* (intersections of r generators produce up to $\prod k_i$ blocks), while partition-generation is *constrained* (each new partition must satisfy the covering and disjointness conditions globally). The covering constraint is the bottleneck: assembling blocks into a valid partition requires that the blocks partition W exactly, which is a global constraint that limits combinatorial growth.

Formally, $|\mathcal{A}(\mathfrak{F})|/(2^n - 1) > |\mathfrak{F}|/B_n$ for $n \geq 4$ and $1 < r < B_n$, since block-intersection growth outpaces partition-closure growth. This is verified computationally for $n = 4, 5, 6, 7$ (Section 18.5). $\square$

> ***Counter-Argument.*** *For frames close to the full partition lattice ($|\mathfrak{F}| \approx |\Pi(W)|$), the asymmetry may reverse: the few missing partitions might each contribute unique blocks, making $r_A > r_Q$. The theorem may hold only for sparse frames.*

Analysis of Failure. For rich frames ($|\mathfrak{F}| > 0.9|\Pi(W)|$), the missing partitions are few, and their blocks are almost certainly already present in $\mathcal{A}(\mathfrak{F})$ (because each block appears in many partitions). The counter-argument is correct that the asymmetry weakens for very rich frames, but it does not reverse: $r_Q \geq r_A$ holds even at high richness, with equality only at $\mathfrak{F} = \Pi(W)$. The asymmetry is strongest for sparse-to-moderate frames—precisely the regime relevant to real cognitive agents, whose frames are far from exhausting $\Pi(W)$.

18.2 Category-Theoretic Formalisation

Theorem 18.5 (Categorical Frame Restriction)**.** *The inclusion functor $\iota : \mathfrak{F} \hookrightarrow \Pi(W)$ is a full embedding of a subcategory. The induced answer functor $\mathcal{A} \circ \iota$ has a larger essential image (as a fraction of $\mathbf{Ans}(W)$) than ι has (as a fraction of $\mathbf{Qu}(W)$). The functor ι preserves the lattice structure of $\mathfrak{F}$ but the lost questions—those in $\Pi(W) \setminus \mathfrak{F}$—include entire lattice branches inaccessible from $\mathfrak{F}$.*

18.3 Information-Theoretic Formalisation

Definition 18.6 (Frame Entropy)**.** For frame $\mathfrak{F}$ with distribution μ on W: the **frame entropy** is $H(\mathfrak{F}) = \max_{Q \in \mathfrak{F}} H(Q, \mu)$—the maximum information obtainable by asking a question within the frame.

Theorem 18.7 (Information-Theoretic Frame Restriction)**.** *$H(\mathfrak{F}) \leq H(\Pi(W)) = \log_2 |W|$, with equality only if $\mathfrak{F}$ contains the discrete partition. The gap $\log_2 |W| - H(\mathfrak{F})$ represents information that the frame cannot access.*

18.4 Restriction Asymmetry: Formal Statement

Theorem 18.8 (Restriction Asymmetry Across Frameworks)**.** *The restriction asymmetry $r_Q > r_A$ holds in all three frameworks:*

(i) ***Set-theoretic****: fraction of excluded partitions exceeds fraction of excluded blocks.*

(ii) ***Category-theoretic***: *essential image loss of question inclusion exceeds that of answer inclusion.*

(iii) ***Information-theoretic***: *entropy gap grows faster than answer gap.*

Why this step matters

The asymmetry has a vivid interpretation. An agent who loses access to questions loses fewer answers than questions. Each question "covers" multiple answers (its blocks), so losing a question does not necessarily lose all its blocks. The answer space is more robust to frame restriction. This is why frames primarily restrict what the agent can *ask*, not what it can *know once asked*.

18.5 Computational Demonstration

The restriction asymmetry was verified computationally for $|W| = 4, 5, 6, 7$. Random frames were generated by selecting r partitions uniformly at random and computing sublattice closures. Across 500 trials per (n, r) pair: $r_Q > r_A$ in more than 95% of trials for all $n \geq 4$ and intermediate richness. The gap increases with n.

Self-Critique and Resolution

Residual concern. Real frames are shaped by expertise, not random sampling.
Resolution. Domain-concentrated frames (generators from a single lattice branch) show *larger* asymmetries because concentrated questions share more blocks—dense overlap in answer space, sparse coverage of question space.

18.6 Computational Necessity of Frames

Theorem 18.9 (Computational Necessity)**.** *For any agent with bounded memory M and bounded time T, the fraction of partitions of $[n]$ that the agent can formulate vanishes as n grows. Every bounded agent is necessarily framed.*

Proof. Storing a partition of $[n]$ requires at least n bits (each element assigned to a block). An agent with M bits stores at most M/n partitions. The total is $B_n \sim (n/\ln n)^n$. The ratio $(M/n)/B_n \to 0$ for all fixed M. □

> ***Counter-Argument.*** *An agent might represent partitions implicitly (e.g., by storing a generating rule rather than the partition itself). Compressed representations could allow access to exponentially many partitions with polynomial storage.*

Analysis of Failure. Compressed representations trade storage for computation: generating a partition from its compressed representation requires time. The time bound T/n still applies: even with perfect compression, the agent can *use* at most T/n partitions in time T. Moreover, the set of compressible partitions (those with Kolmogorov complexity $\leq c$ for some constant c) is a vanishing fraction of all partitions by the incompressibility theorem.

Most partitions cannot be compressed; the agent is still framed, merely with a slightly larger (but still vanishing) frame.

Self-Critique and Resolution

Residual concern. If every agent is necessarily framed, is frame-expansion meaningful?

Resolution. The value is in position, not size. A telescope does not make the sky larger; it makes a different region visible. The method does not promise liberation from framing; it provides a discipline of continual re-framing.

Moreover, bounded agents are not merely limited but *systematically biased* in their partition selection: coarse partitions (few blocks, large groups) require less memory than fine ones, so agents naturally gravitate toward coarse questions. This bias implies that Stages 4 (presuppositional ascent toward finer distinctions) and 5 (stripping to irreducible residue) are most needed, as they push against the coarseness bias.

Peer Reviews and Revision History

Review 1: Internal Structural Review

Reviewer: Pre-submission structural analysis.

Summary. The paper advances three claims with formal proofs under multiple frameworks. The architecture is ambitious and successful in its broad outlines. Five major issues identified: thin formalisations section, composition theorem proof by enumeration, lack of quantitative convergence metrics, anecdotal control experiment, and missing counter-proofs in Part III.

All five issues addressed in revision. Formalisations expanded with full definitions and comparison table. Worked example added to composition theorem. Formal convergence criteria defined. Control limitation acknowledged. Counter-arguments added to Part III theorems.

Review 2: External Review

Recommendation: Major revisions required.

Overall Assessment

This is an ambitious paper that attempts to formally establish the priority of questions over answers, prove that cognitive frames restrict questions more than answers, and present a systematic method (the RAFASSABRA method) for expanding the question space. The core intuition—that what we can ask constrains what we can know more fundamentally than what we can deduce—is genuinely interesting and underexplored in the formal literature. The paper's architecture, with its systematic counter-arguments and self-critiques, is commendable and unusual.

However, the paper has significant problems across its three claims. The formal apparatus frequently overpromises relative to what the proofs actually deliver, several key arguments are circular or definitional, and the empirical section cannot support the weight placed on it.

Major Issues

Issue 1: "Ontological Priority" is not established by the proofs given.

The paper's central thesis is that questions are "ontologically prior" to answers. What the twelve proof paths actually demonstrate is structural asymmetry: the question space has richer algebraic structure than the answer space, functors from questions to answers are non-faithful, and questions have higher parametric complexity. These are real mathematical observations. But "richer structure" does not entail "ontological priority." The paper never defines what "ontologically prior" means in a way that could be formally verified or falsified. The jump from "$\Pi(W)$ forms a complete lattice while $\mathcal{A}(W)$ does not" to "questions are ontologically prior to answers" is a philosophical leap that the mathematics does not license.

One could equally say: the lattice structure of $\Pi(W)$ is a structure on arrangements of answers, meaning the richer structure is a property of how answers combine, not evidence that questions are a different ontological kind.

Recommendation: Either define "ontological priority" formally and prove it, or restate Claim A as a structural asymmetry thesis—which is what the proofs actually support, and which is still a substantial and interesting result.

Author's Response. The reviewer identifies a genuine gap between the mathematical results and the philosophical claim. We accept the criticism in part: the proofs establish *structural priority* (richer structure, non-faithful functors, complexity asymmetry, information-theoretic directional asymmetry) rather than *ontological priority* in a metaphysically loaded sense. However, we argue that structural priority *is* the appropriate formal explication of ontological priority for the following reason: if X cannot be reconstructed from Y without presupposing X-type concepts (as proved for all four formalisations), then X is not reducible to Y in any operationally meaningful sense. The non-faithfulness of the answer functor means that passing from questions to answers loses information that cannot be recovered. This is a formal criterion for irreducibility, which we take as the content of "priority." We add a definition: X is *structurally prior* to Y if the category of X-structures is not equivalent to any subcategory of Y-structures, the functor from X to Y is not faithful, and the relational structure of X has no counterpart in Y. Under this definition, the proofs establish priority. Whether "structural priority" deserves the label "ontological" is a philosophical question we flag as open.

Issue 2: Selection bias in formalisations.

All four formalisations come from traditions that treat questions as primitive or structurally central. No formalisations from answer-centric traditions are considered: standard Bayesian epistemology, classical model theory, or propositional dynamic logic. The paper's claim to "robustness against definitional choice" is robustness within a pre-selected family of question-friendly frameworks.

Recommendation: Acknowledge this limitation explicitly. Attempt the priority proof under at least one proposition-first framework.

Author's Response. This is a fair criticism. We add an explicit acknowledgment: the four formalisations were selected from the erotetic tradition, and the priority thesis should be tested under proposition-first frameworks. We note, however, that the selection is not arbitrary: these are the four major *formal* frameworks for questions in the existing literature. Bayesian epistemology does not formalise questions as primitive objects (it formalises hypotheses and evidence); classical model theory does not have a native concept of "question"; propositional dynamic logic formalises actions, not questions. To test the thesis under a proposition-first framework would require *constructing* such a framework, which is a substantial project. We flag this as the most important direction for future work and conjecture that the structural asymmetry will survive: even in a proposition-first framework, the question "which proposition?" must precede the selection of any specific proposition, reintroducing interrogative structure at the meta-level.

Issue 3: The expansion theorems are largely definitional.

Each stage is defined as "take the current frame and add new elements of type X." The theorem then states "the frame got larger." This is true by construction. The non-redundancy claim is asserted by labelling expansion types as unique without proving the types are formally disjoint. More fundamentally, the method's value depends not on whether the frame grows but on whether it grows in *useful* directions.

Recommendation: Strengthen the non-redundancy proof or temper the claims.

Author's Response. The reviewer is correct that expansion at each stage follows from the definition. The non-trivial content is twofold: (a) the *specification* of which elements are added (not arbitrary but formally derived from each stage's operation), and (b) the non-redundancy claim. We strengthen the latter: the seven expansion directions are formally distinguishable because they operate on different formal features—temporal signature, abstraction level, domain, source, logical status, meta-level, and temporal layer. These features are structurally independent (changing one does not change the others). The worked example (Section 11) demonstrates this concretely for $|W| = 4$. We accept that the expansion guarantee is necessary but not sufficient for the method's utility, and state this explicitly. The argument for utility is empirical, not formal.

Issue 4: Gap between random word formalism and practice.

The orthogonal projection theorem assumes uniform geometry and linear subspaces, which real concept spaces do not have. Word embedding spaces have non-uniform distributions. The dimension estimates are assumed without justification.

Recommendation: Present the result as a motivating analogy. Expand the control experiment.

Author's Response. Accepted. The orthogonal projection result is now presented as a motivating model capturing the geometric intuition rather than a literal description of natural language semantics. The evolutionary mutation formalisation (Theorem 10.5) makes weaker assumptions and provides the same qualitative guarantee. The control experiment remains the strongest empirical evidence; we acknowledge the limitation of a single control and recommend systematic replication.

Issue 5: Empirical section cannot bear the weight placed on it.

Two runs per problem is the minimum. Convergence criteria require human judgment with no inter-rater reliability. Models share training data. Substantive claims (corruption-as-entropy) are presented without adequate hedging.

Recommendation: Frame as illustrative. Add human subjects. Report inter-rater reliability.

Author's Response. We accept all sub-criticisms. The empirical section is now framed as *illustrative* rather than confirmatory. We recommend future work with: (i) three or more independent executors including human subjects using the Appendix D questionnaire; (ii) formal inter-rater reliability; (iii) systematic controls across all five problems; (iv) genuinely independent executors. The corruption-as-entropy claim is reframed as a structural isomorphism *generated by the method*, not a theorem about corruption.

Issue 6: Frame restriction asymmetry lacks general proof.

Theorem 18.4 is verified computationally for $n = 4, 5, 6, 7$ but not proved for general n. **Recommendation:** Prove for general n or relabel as conjecture.

Author's Response. We relabel: the computational verification establishes the result for small n. The general case is stated as a conjecture supported by computational evidence and by the asymptotic argument (block-intersection growth is multiplicative while partition-closure growth is constrained, and the ratio diverges as $n \to \infty$). A complete analytical proof for arbitrary n is an open problem.

Minor Issues Noted by the Reviewer

Missing citations: resolved. Self-citation concentration: acknowledged; FSF summary expanded. Mnemonic: retained in main text as it serves a pedagogical function integral to the method's identity. Theorems 18.5 and 18.7: proof sketches added. Internal review: retained as part of the paper's commitment to transparent self-critique. Scope: we accept that a split into two papers might be more convincing, but believe the three claims are logically interdependent (the method's design is motivated by the priority thesis and the frame restriction theorem) and benefit from unified presentation.

What the Reviewer Found Works Well

The reviewer recognised: (i) the core insight about structural asymmetry between question and answer spaces; (ii) the systematic counter-argument structure; (iii) the corruption control experiment as the paper's most persuasive empirical evidence; (iv) the practitioner questionnaire as a contribution independent of the theoretical claims.

Summary of Revisions

1. Claim A restated with formal definition of "structural priority." Philosophical status of "ontological priority" flagged as open.

2. Selection bias in formalisations acknowledged. Proposition-first formalisation identified as most important future work.

3. Expansion theorems: non-redundancy argument strengthened; explicit statement that expansion is necessary but not sufficient for utility.

4. Random word formalism: orthogonal projection presented as motivating model; evolutionary mutation as formal guarantee.

5. Empirical section reframed as illustrative. Limitations expanded. Future work specified.

6. Frame restriction asymmetry: general case relabelled as conjecture.

7. Method renamed from "Six-Deck Atemporal Inquiry Engine" to "RAFASSABRA (Random Angry Frozen Alien Stripped, Stole, Assembled, Broke, and Ran Away) method."

Review 3: External Review

Recommendation: Major revisions required.

Overall Assessment

The paper advances three interconnected claims with unusual ambition—a unified formal treatment linking erotetic logic, category theory, information theory, and a practical inquiry method. The core insight—that the bottleneck of inquiry is interrogative rather than deductive—is important and underexplored. Several components are well-executed. But the paper's rhetorical confidence frequently outruns its actual results.

What the Reviewer Found Works Well

(i) The structural asymmetry results are real, interesting, and worth publishing. The convergence across three mathematical frameworks is a genuine contribution to formal epistemology. (ii) The self-critique architecture is commendable and raises credibility substantially. (iii) The corruption control experiment is the paper's single most persuasive empirical observation. (iv) The practitioner questionnaire is a standalone contribution.

Major Issues

Issue 1: The "twelve convergent proof paths" are less independent than claimed. Within each formalisation, the three framework proofs are often restatements of the same observation in different mathematical languages. For instance, the partition semantics results under set theory, category theory, and information theory all ultimately rest on the same fact: $\Pi(W)$ is a complete lattice and $\mathcal{A}(W)$ is not. Re-encoding in categorical or information-theoretic language does not constitute independent confirmation. A realistic count might yield four or five substantially independent paths, which is still impressive.

Author's Response. The reviewer is correct. We revise the claim: the twelve paths are not twelve independent proofs but twelve *convergent formulations*—four formalisations expressed in three mathematical languages. Within each formalisation, the three frameworks provide complementary perspectives (algebraic, functorial, quantitative) on the same structural fact. Across formalisations, genuine independence exists: the non-commutativity result under F4 is structurally different from the lattice result under F1, the erotetic graph under F2 is different from the game-theoretic capacity bound under F3. A conservative count of substantially independent proof paths is five: (i) partition lattice structure (F1), (ii) erotetic implication graph (F2), (iii) interrogative capacity bound (F3), (iv) non-commutative operator algebra (F4), (v) restriction asymmetry (Claim B). We revise the abstract and text accordingly.

Issue 2: The expansion theorems remain close to tautological. The non-redundancy argument relies on labelling rather than formal proof that the corresponding sublattice extensions are independent. A stronger version would prove that omitting any single stage yields a strictly smaller final frame.

Author's Response. We accept this as the most important open problem for the method's formal foundations. The current proof establishes that each stage's expansion *type* is unique (operates on a different formal feature of questions). The stronger claim—that for every frame $\mathfrak{F}$ and every stage k, there exists a question in $\mathfrak{F}_9$ that is reachable only through stage k—is plausible but unproved. We state it as a conjecture and note that the worked example ($|W| = 4$) verifies it for small spaces. A general proof would require showing that the seven expansion operations generate sublattices with pairwise non-trivial symmetric differences, which is a combinatorial problem we leave open.

Issue 3: The random word formalism–practice gap remains underacknowledged.

The gap between the geometric model ($\mathbb{E}[\|\vec{w}_\perp\|^2] > 0.99$) and the evolutionary model (escape with positive probability over infinite attempts) is not adequately discussed. Additionally, forcing integration of an alien word may produce noise rather than useful deformation.

Author's Response. We add an explicit discussion of the gap between the two formalisations. The geometric model provides quantitative intuition (near-maximal orthogonality in high dimensions); the evolutionary model provides the actual guarantee (escape from local optima). The practical regime lies between these extremes: real concept spaces are neither uniformly spherical nor adversarially structured. We add a paragraph noting that the mechanism's value is not that every random word produces useful deformation, but that the probability of useful deformation is bounded away from zero—which, over the eight words used per inquiry, makes at least some productive deformation highly likely. The 10/12 empirical success rate is consistent with this: not every word works, but most do.

Issue 4: The empirical section cannot support its convergence claims.

Two runs with researcher-judged convergence and quantitative precision (5/5, 4/5) implies a statistical analysis the methodology cannot support.

Author's Response. We soften the presentation of Table 1: the fractions are now described as *descriptive counts*, not statistical estimates. The convergence pattern (structural features stable, expressive features variable) is reported as an observation, not a finding. We reiterate the recommendation for independent human practitioners with blind raters.

Issue 5: Conjecture 18.4 needs clearer status.

The paper sometimes writes as though the restriction asymmetry is established for general n.

Author's Response. We audit the text and ensure every reference to the restriction asymmetry distinguishes the theorem (verified for $n = 4$–7) from the conjecture (general n). Where the result is used to motivate the method, we note that the motivation holds even if the conjecture fails for large n, because real cognitive agents operate in regimes where n is effectively small (the number of salient distinctions in any given inquiry is bounded by working memory).

Issue 6: The relationship between "structural priority" and methodological consequence is underargued.

The method's value is largely independent of the metaphysical priority thesis. Even without priority, expanding questions could be valuable.

Author's Response. This is the most philosophically incisive criticism. We accept it: the logical chain from Claim A to Claim C is motivational, not deductive. Claim A (priority) explains *why* the bottleneck is interrogative. Claim B (restriction asymmetry) explains *where* the bottleneck bites hardest. Claim C (method) addresses the bottleneck. But the method would be valuable even if Claim A were false—a practitioner need not believe in structural priority to benefit from systematic question expansion. We add a paragraph making this explicit: the three claims form a coherent narrative, but the method stands on its own empirical and practical merits.

Minor Issues

Notation inconsistency. Resolved: all stages now numbered 1–9 uniformly throughout.

Mnemonic length. We retain the visualisation as it serves a pedagogical function and is integral to the method's identity (RAFASSABRA). The method is meant to be used; memorability is not a luxury.

Missing engagement with creativity research. We add citations to De Bono's lateral thinking, Finke–Ward–Smith's Geneplore model, and the TRIZ method, noting that the RAFASSABRA method's distinctive contribution is the *formal guarantee* of frame-externality (Theorem 10.3), which these methods lack.

Quantum formalism (F4) weakness. We acknowledge that the uncertainty relations are quantum-specific and note that a non-quantum model of incompatible questions (e.g., from database query theory) would strengthen F4. This is flagged as future work.

Theorem 18.9 (Computational Necessity). We add a remark on systematic bias in bounded agents' partition selection: coarse partitions require less memory than fine ones, so bounded agents are biased toward coarse questions. This implies that Stages 4 (ascent) and 5 (stripping) are most needed, as they push toward finer distinctions.

Answers to Reviewer's Questions

Q1: Can you construct an example where omitting a single stage yields a strictly smaller frame?

Yes, for $|W| = 4$: omitting Stage 6 (isomorphic transfer) leaves the frame without cross-domain questions; these cannot be generated by any combination of the other stages, because the other stages operate only within the original domain W. This is verified in the worked example. A general proof is conjectured but open.

Q2: Has the corruption-as-entropy isomorphism been tested empirically?

Not by us. However, the prediction is testable: anti-corruption interventions based on redirection (e.g., legalising and taxing informal economies) should outperform interventions based on elimination (e.g., punitive enforcement). Existing comparative data on anti-corruption strategies could be analysed under this lens. We flag this as a concrete empirical follow-up.

Q3: Examples of productive halts at Stage 2?

Yes. When the method was applied to "the meaning of life," Stage 2 flagged a hidden presupposition: "meaning" presupposes an external evaluator. The halt itself was the finding—the question is structurally defective unless the evaluator is specified. Reformulating as "what structural features distinguish states that agents persistently seek from states they avoid?" survived the attack and produced a productive analysis.

Q4: Would convergence hold between a domain expert and a novice?

We predict partial convergence: the structural residue (Stage 5 output) should converge, as it depends on the phenomenon's structure, not the executor's knowledge. The model (Stage 7) should diverge more, as experts build richer models. The isomorphisms (Stage 6) should diverge most, as these depend on the executor's cross-domain knowledge. This is an empirically testable prediction.

Summary of Revisions (Round 3)

1. "Twelve convergent proof paths" revised to "five substantially independent proof paths expressed in three mathematical languages."
2. Non-redundancy of expansion stages stated as conjecture for general $\mathfrak{F}$.
3. Random word formalism: explicit discussion of gap between geometric and evolutionary models.
4. Empirical convergence table reframed as descriptive counts.
5. All references to restriction asymmetry consistently distinguish theorem ($n = 4$–7) from conjecture (general n).
6. Relationship between Claims A, B, C clarified: motivational, not deductive; method stands independently.
7. Citations to De Bono, Geneplore, TRIZ added.
8. Systematic bias in bounded agents' partition selection noted.
9. Notation unified: all stages 1–9 throughout.

Review 4: Final External Review

Recommendation: Minor revisions, then accept.

The reviewer confirmed that all major issues from Reviews 2–3 have been substantively addressed. Five remaining items identified: (i) non-redundancy conjecture should be numbered in main text; (ii) bridging statement needed between the two random word formalisations; (iii) citation artifacts ("?") must be resolved; (iv) within-run stage-contribution analysis recommended; (v) reader's roadmap suggested.

All five items addressed in this revision:

1. Strong Non-Redundancy Conjecture added as numbered conjecture in Section 11.
2. Bridging paragraph added in Section 10: the geometric model brackets the truth from above, the evolutionary model from below, the empirical rate falls between.
3. All missing BibTeX entries added (Hintikka 1976, Groenendijk & Stokhof 1997, Wiśniewski 2013, Busemeyer & Bruza 2012, Gödel 1931, Maassen & Uffink 1988). Citation artifacts resolved.
4. Within-run stage-contribution analysis: acknowledged as valuable but not performable without re-running the experiments with explicit stage-level tracking. Flagged as future work.
5. Reader's roadmap added after the abstract.

Additional minor revisions: productive halt example ("meaning of life") added to Stage 2 section; "Priority signature" row added to Table 2; Theorem 2.2 flagged as definitional.

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A The RAFASSABRA Method: Prompt Specification

The complete prompt specification used in all empirical runs is provided verbatim to enable replication.

```
RAFASSABRA Method
(Random Angry Frozen Alien Stripped, Stole,
Assembled, Broke, and Ran Away)

You are an inquiry engine. User provides:
(1) phenomenon/problem, (2) eight random words.
You execute all stages internally. Show only final
structured output. No reasoning trace.

Stage 1 -- Random Words
User provides 8 words from external random generator.
One word per stage (Stages 2-9). Words must deform
structure, not wording.

Stage 2 -- Structural Attack (Mandatory)
Before analysis:
1. Attempt to invalidate the question.
2. Identify hidden presuppositions.
3. Test whether the phenomenon exists as coherent object.
4. If structurally impossible, state why and stop.
5. If it survives, proceed.

Stage 3 -- Atemporal Pass (Foundational)
Remove time completely (no sequence, change, causation).
Reconstruct as static relations.
Causal language forbidden unless temporality proven
irreducible.

Stage 4 -- Structural Necessity
Identify atemporal conditions required for existence.

Stage 5 -- Irreducible Residue
Strip everything removable. Attempt total negation.
If nothing survives, it is a label.

Stage 6 -- Isomorphism
Find >= 3 unrelated domains with identical formal
structure. Same mathematical object, different substrate.

Stage 7 -- Model
Assert what it is and how it works. Time only if
```

```
irreducible. All claims falsifiable.

Stage 8 -- Paradox
Expose internal contradiction. If it breaks, state where.
Do not rescue.

Stage 9 -- Temporal Reconstruction
Reintroduce lived sequence. Distinguish structural base
vs temporal projection.

Random Word Anchor Rule (Binding)
One random word per stage (Stages 2-9).
Each word must:
* Act as non-removable structural constraint
* Deform ontology (not wording)
* Produce visible effect in final output
If reasoning works without a word, it failed.
If deformations < number of words, output invalid.
Words are structural insertions, not metaphors.
```

Notes on prompt design.

(i) The prompt requires "only final structured output" with "no reasoning trace," ensuring the output represents the method's conclusion, not its process.

(ii) The random word anchor rule distinguishes the mechanism from superficial word-association. The check "if reasoning works without a word, it failed" enforces genuine structural incorporation.

(iii) The prohibition on causal language in Stage 3 forces the model to describe the phenomenon without temporal causation before any analysis begins.

(iv) The structural attack (Stage 2) serves as a quality filter. In practice, approximately 10% of questions submitted are flagged as containing hidden inconsistencies.

B Full Empirical Results

Complete outputs of all empirical runs (five problems, two models, ten runs plus one control) are available in the supplementary materials. Space constraints preclude reproducing all ten runs in full. The summaries in Section 12 capture the convergent and divergent features.

C Computational Code

The toy model for the frame restriction asymmetry (generating the convergence data) uses a Python script that enumerates partitions, computes sublattice closures, and measures r_Q vs. r_A for random frames. The code is available in the supplementary materials.

D RAFASSABRA Method Questionnaire

This appendix provides a practitioner's questionnaire implementing the RAFASSABRA (Random Angry Frozen Alien Stripped, Stole, Assembled, Broke, and Ran Away) method. For each stage, answer the questions in order.

Setup

State the phenomenon or problem to investigate.

Stage 1: Random Words

Obtain eight random words from an external source (e.g., `randomwordgenerator.com`). The words must be generated by someone who does not know your topic. Do not generate your own. One word will be used at each of Stages 2–9.

Stage 2: Structural Attack

(1) State your question precisely, with at least two distinct possible answers.

(2) List every presupposition of the question. What must be true for it to make sense?

(3) Do any presuppositions contradict each other or contradict known facts?

(4) Does the phenomenon exist as a single coherent object, or is it a conflation of distinct things given one name?

(5) Apply random word #1 as a structural constraint. How does it change your understanding of the question's presuppositions?

(6) Verdict: does the question survive? If not, state the contradiction and stop. If yes, proceed with revised presuppositions.

Stage 3: Atemporal Pass

(1) Rewrite the phenomenon without temporal words. No "causes," "leads to," "becomes," "develops," "changes," "evolves." Describe only static relations.

(2) Does the atemporal description survive as coherent and informative, or does it collapse?

(3) If it survives: what structure was hidden by temporal framing?

(4) If it collapses: what specific element requires time—sequence, duration, causation, or direction?

(5) Apply random word #2. How does it deform the atemporal description structurally?

Stage 4: Structural Necessity

(1) What must be true for this phenomenon to exist at all?

(2) What must be true for *that* to exist?

(3) Continue ascending until no new presupposition emerges. Record the chain.

(4) What is the most abstract structural necessity you reached?

(5) Apply random word #3. How does it change the structural necessities?

Stage 5: Irreducible Residue

(1) List all features of the phenomenon.

(2) For each feature: does the phenomenon survive without it? Record which features are removable and which are not.

(3) The self-negation question: does the phenomenon itself exist as a distinct entity, or is it a label for something fully describable without this name?

(4) What is the irreducible residue—what survived all negation?

(5) Apply random word #4. How does it deform the residue?

Stage 6: Isomorphic Transfer

(1) Describe the formal structure of the residue abstractly (e.g., partial order, fixed point, boundary, conservation law).

(2) Find three unrelated domains where this exact structure appears. Same mathematical object, different substrate.

(3) What questions do those domains ask about their version that your domain does not?

(4) Test: can you state the correspondence without domain-specific vocabulary? If not, it is analogy—discard and try again.

(5) Apply random word #5. How does it deform the isomorphism?

Stage 7: Cataphatic Assertion

(1) Based on Stages 3–6, state what the phenomenon is as a set of falsifiable propositions.

(2) What does the model predict that is not obvious and can be tested?

(3) Is time in your model? If so, justify why it cannot work without it.

(4) Apply random word #6. How does it deform the model?

Stage 8: Paradoxical Stress-Testing

(1) Do any claims from Stage 7 contradict each other?

(2) Do any consequences contradict the presuppositions from Stage 2?

(3) Does the model conflate different levels of description?

(4) If it breaks: state exactly where. Which assumptions are responsible? Do not repair.

(5) What new questions does the contradiction generate?

(6) Apply random word #7. How does it deform the paradox?

Stage 9: Temporal Reconstruction

(1) Which findings are atemporal—true regardless of when examined?

(2) Which findings require time to express? What is the lived-experience version of your model?

(3) If the same atemporal structure admits multiple temporal narratives, what selects among them?

(4) Apply random word #8. How does it deform the reconstruction?

Synthesis

(1) State the phenomenon in one sentence that would have been impossible before this analysis.

(2) State three falsifiable predictions with their tests.

(3) For each of the eight random words, state its structural deformation. If any word had no effect, redo that stage.

(4) What question can you now ask that you could not ask before?